大学物理

电子教案

清华大学物理系

第二章 牛顿运动定律

§ 2.1 牛顿运动定律

§ 2.2 SI 单位和量纲（自学）

§ 2.3 常见力

§ 2.4 基本自然力

§ 2.5 应用牛顿定律解题

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§ 2.7 平动参考系的惯性力

§ 2.8 转动参考系的离心力和科氏力

§ 2.9 *潮汐力和潮汐

§2.1 牛顿运动定律

一. 牛顿第一定律（惯性定律）和惯性系

任何物体如果没有力作用在它上面，都将保持静止的或作匀速直线运动的状态。

1. 定义了惯性参考系

静止或运动相对谁?　　惯性系

遥远的星体作为惯性系　　傅科摆

2. 定性了物体的惯性和力

保持运动状态和改变运动状态　　自洽

二. 牛顿第二定律

$$\vec{F} = \lim_{\Delta t \to 0} \frac{\Delta \vec{p}}{\Delta t} = \frac{d\vec{p}}{dt} = m\vec{a}$$

$$\vec{p} \to m\vec{\upsilon}$$

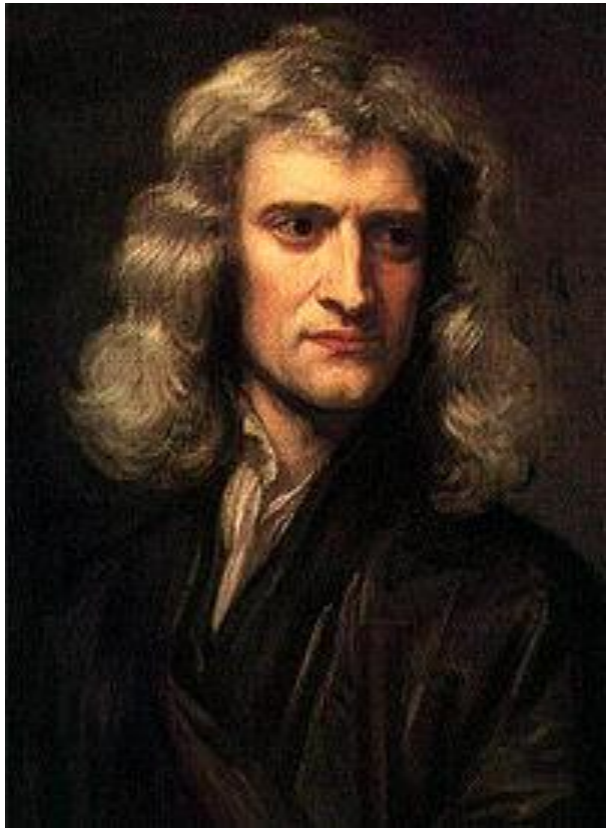
力的叠加原理　　m 为惯性质量

三. 牛顿第三定律(作用力与反作用力)

作用力与反作用力大小相等、方向相反，
作用在不同物体上

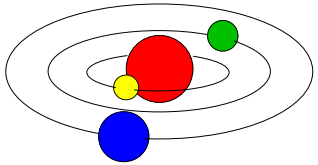
1. 超距作用；2. 电和磁作用下有时不成立.

牛二,三 只在惯性系成立



Nature and nature's laws
lay hid in night,
God said "Let Newton be"
and all was light.

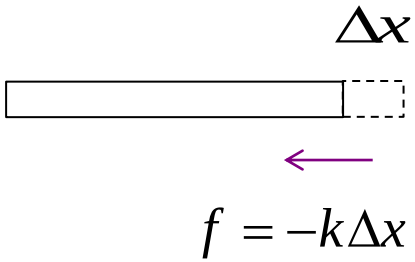
Isaac Newton
25 December 1642 – 20 March 1727

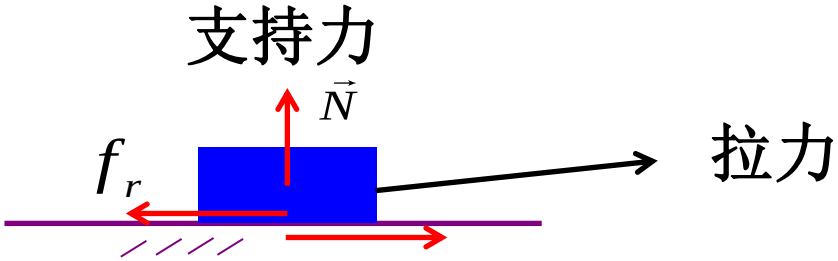


§ 2.3 常见力

1. 重力 mg

2. 弹性力

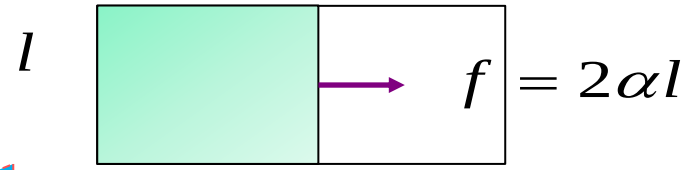




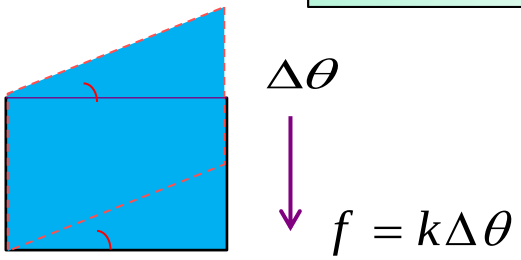
3. 摩擦力 $f_r = \mu N$ $\mu_{滑} < \mu_{最大静}$

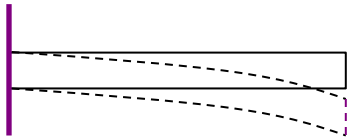
4. 流体阻力(拖曳力) $\vec{f}_r = -(k_1 + k_2 \upsilon)\vec{\upsilon} + \cdots$

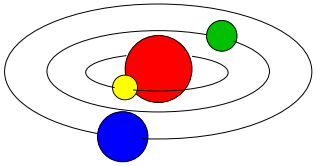
5. 液体表面张力



6. 剪切力







§2.4 基本自然力

自然界有**四种**基本相互作用力：

1：万有引力（时空弯曲效应），如：重力

2：电磁力

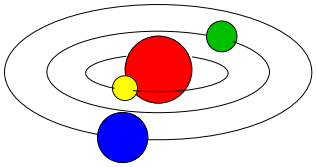
各种**常见力**的来源：弹性力，摩擦力，流体拖曳力，表面张力，剪切（扭曲）力，等

3：弱力，β衰变的原因

4：强力，如核力

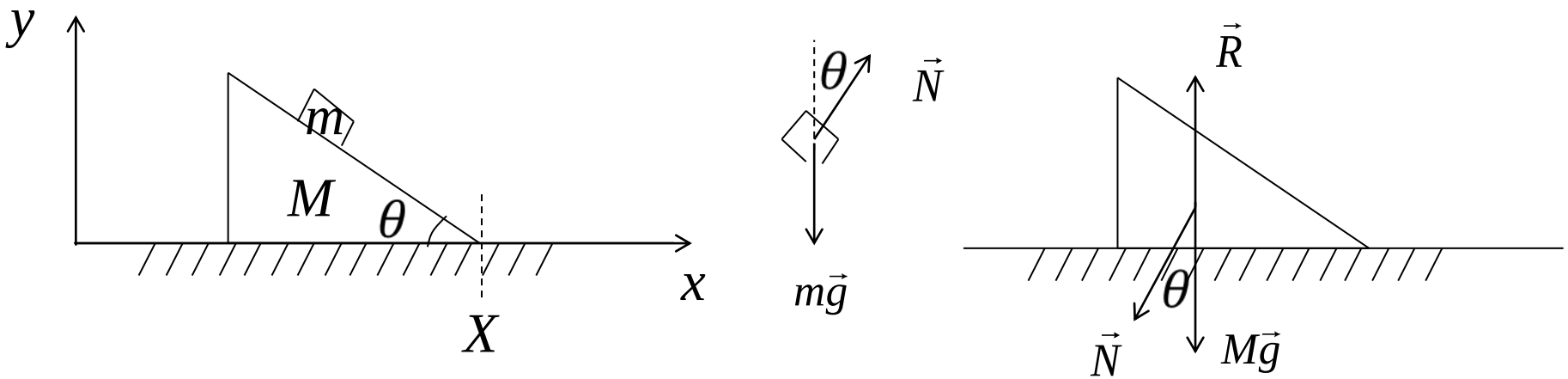
后三个都有**量子理论**

引力是经典理论



§ 2.5 应用牛顿定律解题

例：平面上有一斜面，其上有一物体自由下滑。
忽略所有摩擦，求物体加速度。



解：地面参考系建坐标

隔离受力图（free-body diagram)

$$N\cos\theta - mg = ma_y$$

$$N\sin\theta = ma_x$$

$$-N\sin\theta = MA_x$$

$$R = N\cos\theta + Mg$$

尖:（X，0）

m: (x, y)

$$y = (X - x)\tan\theta$$

$$\ddot{y} = (\ddot{X} - \ddot{x})\tan\theta \quad \rightarrow \quad a_y = (A_x - a_x)\tan\theta$$

$$a_x = \frac{g\sin\theta\cos\theta}{1+\frac{m}{M}\sin^2\theta} \qquad a_y = -\frac{(1+\frac{m}{M})g\sin^2\theta}{1+\frac{m}{M}\sin^2\theta}$$

$$A_x = -\frac{\frac{m}{M}g\sin\theta\cos\theta}{1+\frac{m}{M}\sin^2\theta}$$

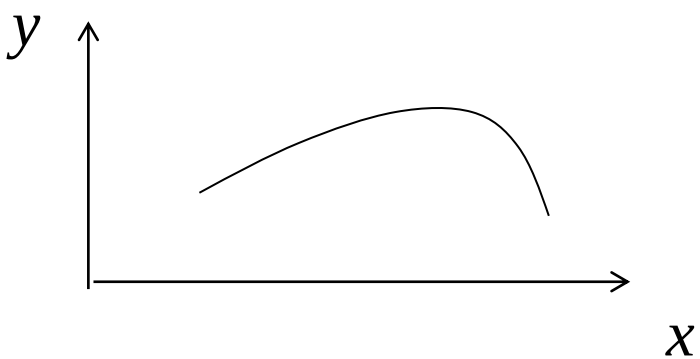
$$M >> m$$

$$a_x = g\sin\theta\cos\theta \qquad a_y = -g\sin^2\theta$$

$$A_x = 0$$

抛体运动

假设运动速度不大，空气阻力(拖曳力) $\vec{f}_r = -k\vec{\upsilon}$

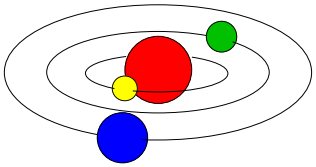


$$-mg\hat{y} - k\vec{\upsilon} = m\vec{a}$$

$$\dot{\vec{\upsilon}} + \beta\vec{\upsilon} + g\hat{y} = 0 \qquad \beta = \frac{k}{m}$$

$$\vec{\upsilon} = -\frac{g}{\beta}\hat{y} + (\vec{\upsilon}_0 + \frac{g}{\beta}\hat{y})e^{-\beta t}$$

$$\vec{r} = \vec{r}_0 - \frac{gt}{\beta}\hat{y} + (\vec{\upsilon}_0 + \frac{g}{\beta}\hat{y})\frac{(1-e^{-\beta t})}{\beta}$$



单选题 1分

设置

只要物体受力, 物体就会加速; 而物体不受力时, 物体就一定静止或做匀速直线运动?

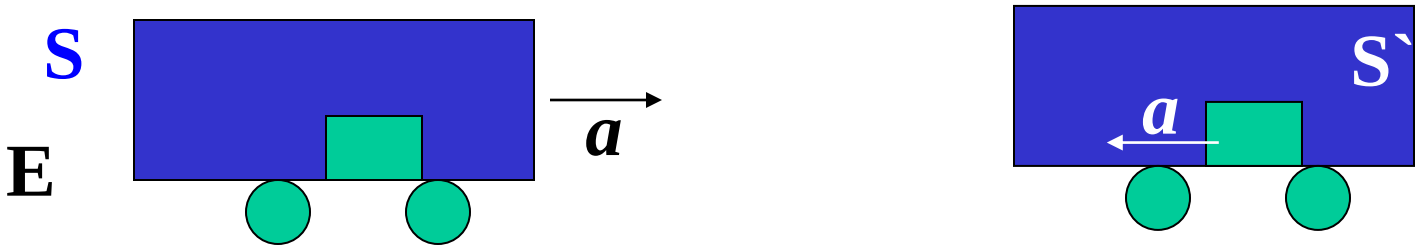
A 是

B 否

C 不一定

提交

§ 2.6 惯性系(Inertial frame)和非惯性系



在 S 参考系，运动符合牛顿定律，在 S` 则不然

牛顿定律在惯性系成立

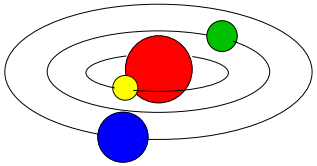
近似惯性系：

地面参考系，自转加速度 $a \sim 3.4\ \mathrm{cm/s^2}$

地心参考系，公转加速度 $a \sim 0.6\ \mathrm{cm/s^2}$

太阳参考系，绕银河系加速度 $a \sim 3 \times 10^{-8}\ \mathrm{cm/s^2}$

MILKY WAY
CORE
10,000
光年
20,000
光年
你在这里
太阳系



§ 2.7 平动参考系的惯性力

设 S 系为惯性系，S ' 系为非惯性系

两个平动参考系之间，加速度变换 $\boxed{\vec{a} = \vec{a}' + \vec{a}_0}$

质点 ***m*** 在 S 系 $\boxed{\vec{F} = m\vec{a}}$ $\vec{F}, m$ 不随参考系变化

在 S ' 系 $\vec{F} \neq m\vec{a}'$ 牛二在非惯性系不成立

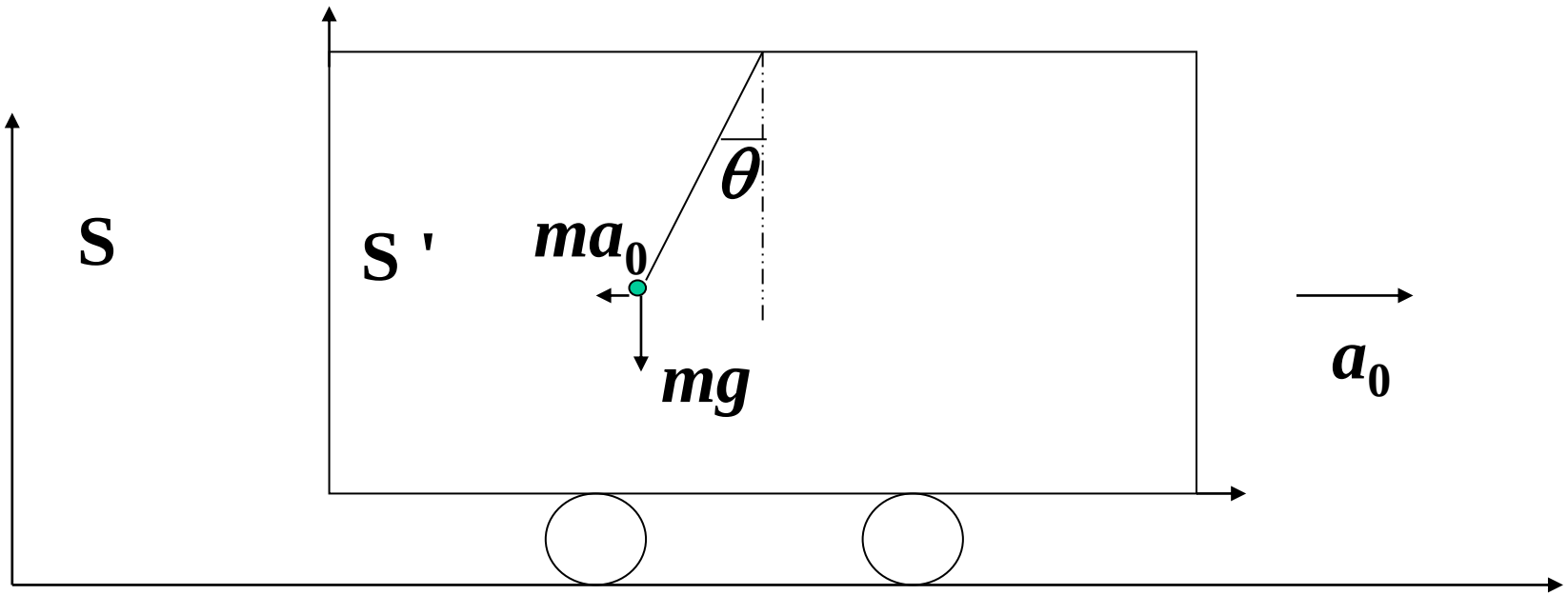
在非惯性系引入虚拟力或惯性力 $\vec{F}_0 = -m\vec{a}_0$

在非惯性系 S '系 $\boxed{\vec{F} + \vec{F}_0 = m\vec{a}'}$

引入惯性力后，牛二在非惯性系形式上成立

例：一匀加速运动的车厢内，观察单摆，平衡位置和振动周期如何变化？

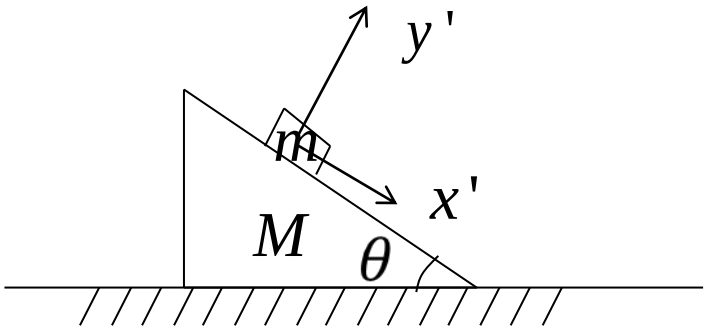
（加速度 a_0，摆长 l，质量 m）

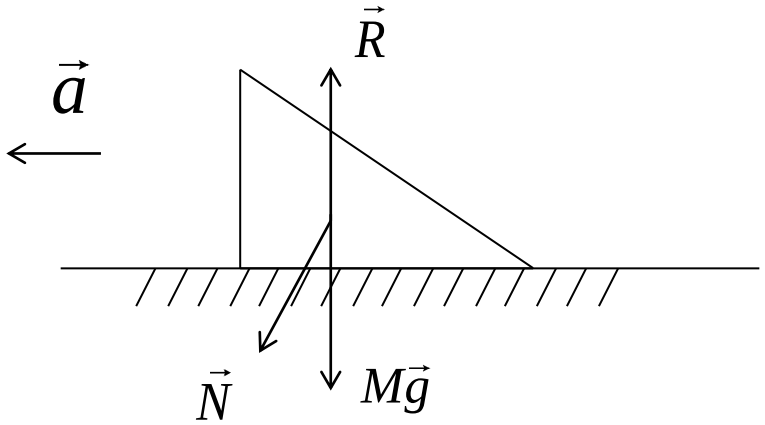


解：在 S '系 $a=\sqrt{a_0{}^2+g^2}$ 平衡位置 $\theta=\tan^{-1}\dfrac{a_0}{g}$

周期 $T=2\pi\sqrt{\dfrac{l}{g}} \rightarrow T=2\pi\sqrt{\dfrac{l}{a}}$

例：平面上有一斜面，其上有一物体自由下滑。
忽略所有摩擦，求下滑物体**相对斜面加速度**。





解：考察斜面运动

$$N\sin\theta = Ma$$

$$R = N\cos\theta + Mg$$

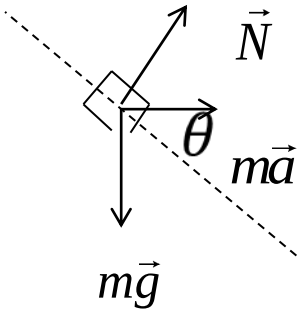
斜面参考系上建坐标

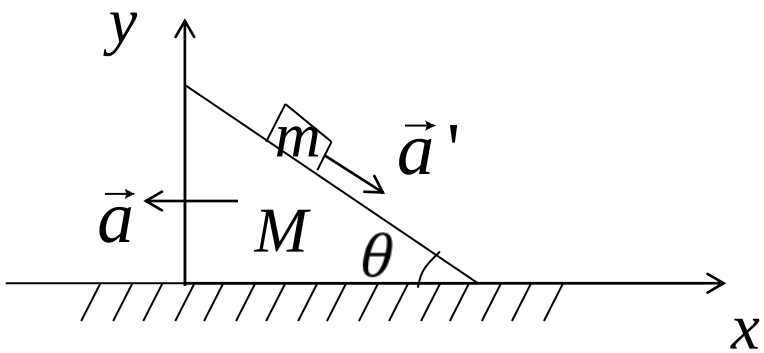
隔离受力图（free-body diagram)

考虑惯性力

$$mg\sin\theta + ma\cos\theta = ma'$$

$$N + ma\sin\theta = mg\cos\theta$$





$$a = \frac{mg\sin\theta\cos\theta}{M + m\sin^2\theta}$$

$$a' = \frac{(1+\frac{m}{M})g\sin\theta}{1+\frac{m}{M}\sin^2\theta}$$

$$M >> m$$

$$a = 0 \qquad a' = g\sin\theta$$

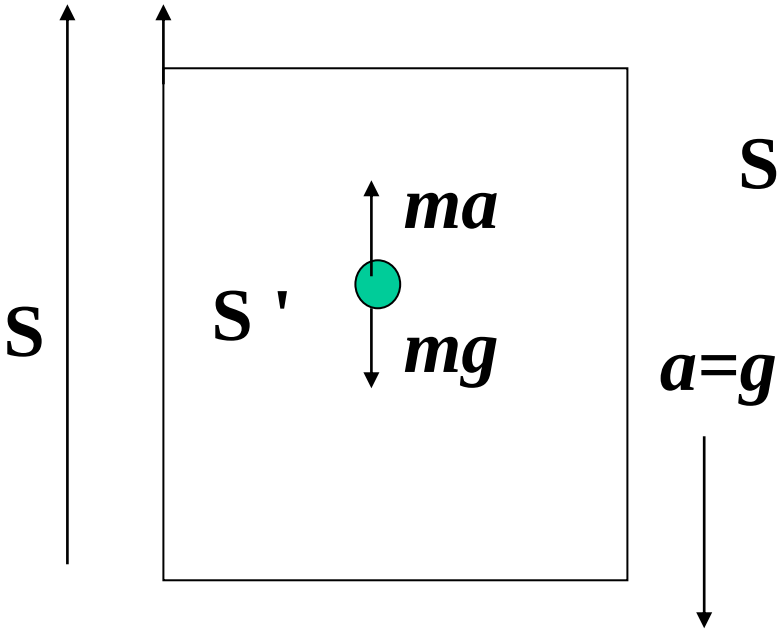
$$\vec{a}_m = \vec{a}\,' + \vec{a}$$

$$a_{mx} = a'\cos\theta - a$$

$$a_{my} = -a'\sin\theta$$

在非惯性系应用牛顿定律，只需考虑惯性力

例：自由落体的参照系



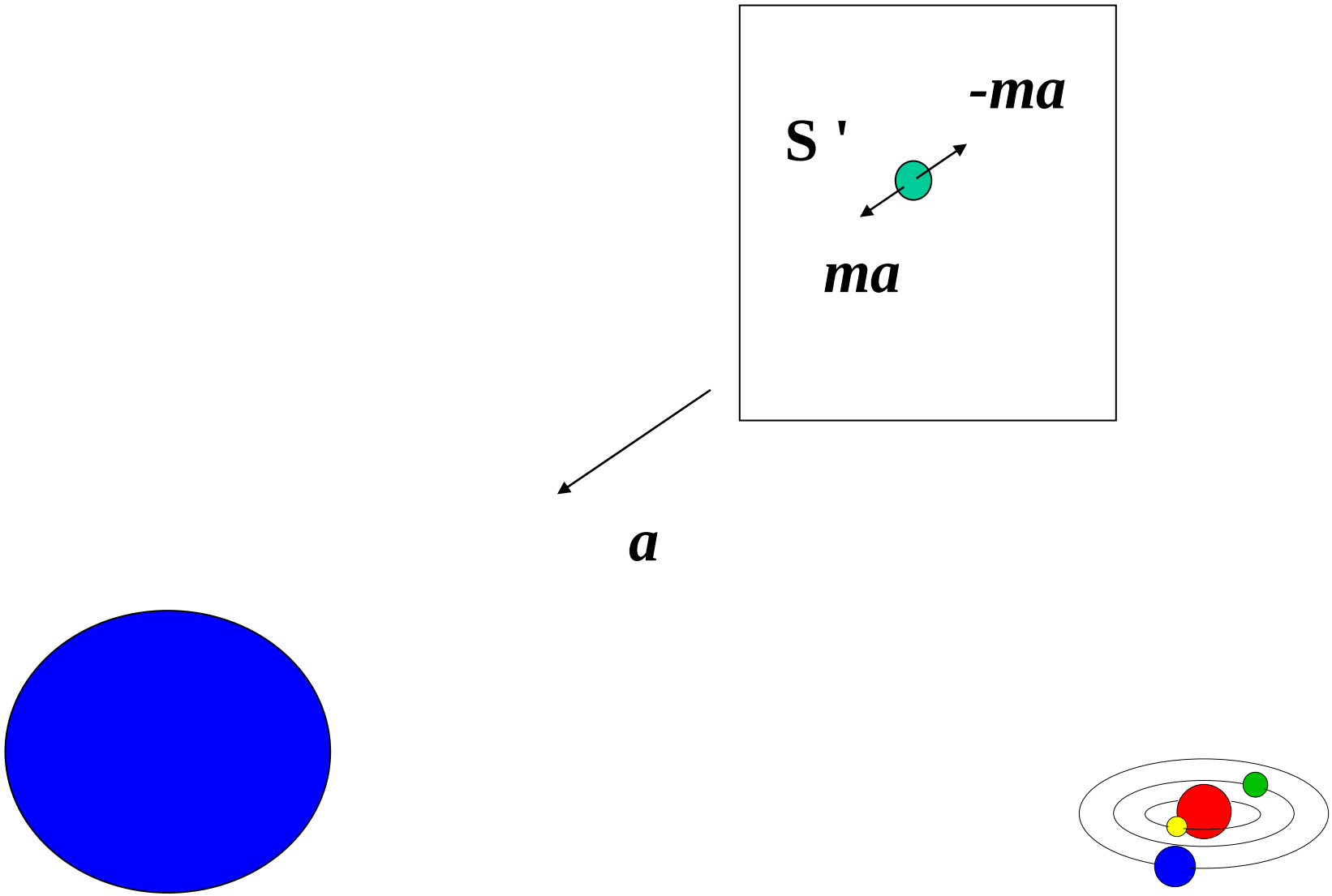
S '是理想的无外力作用的参考系

可以严格检验惯性定律

爱因斯坦的广义相对论

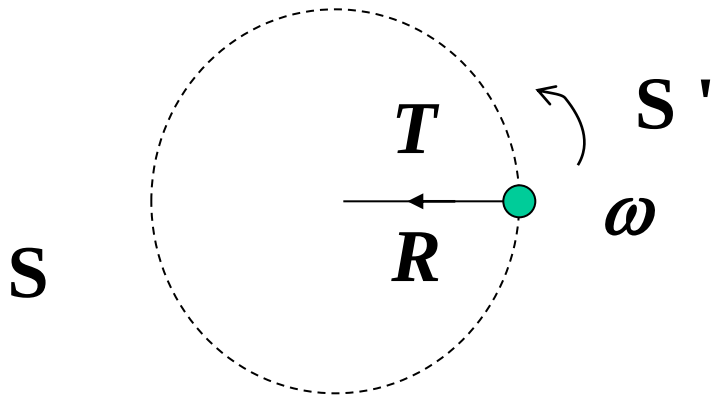
自由落体的参照系可以看作是一个惯性系

例：太空仓（自由落体的参照系）



§ 2.8 转动参考系的离心力和科氏力*

一、离心力



在 S 向心加速度

$$a = R\omega^2$$

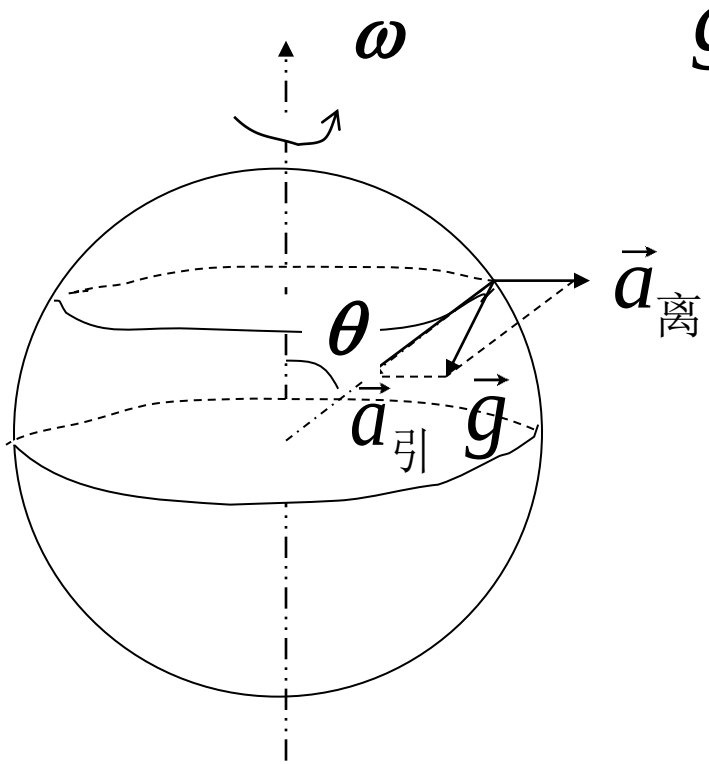
$$T = mR\omega^2$$

质点 m 在 S' 转动参考系静止

$$\vec{T} + \vec{F}_0 = 0 \qquad F_0 = -mR\omega^2$$

离心方向

重力加速度



$$g^2 = a_{引}^2 + a_{离}^2 - 2a_{引}a_{离}\sin\theta$$

$$a_{引} >> a_{离}$$

$$g \approx a_{引} - a_{离}\sin\theta$$

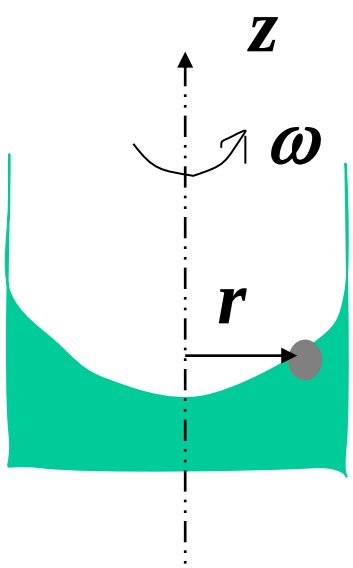
$g_{赤道}$ **= 9.778 m/s²**

$g_{北极}$ **= 9.832 m/s²**

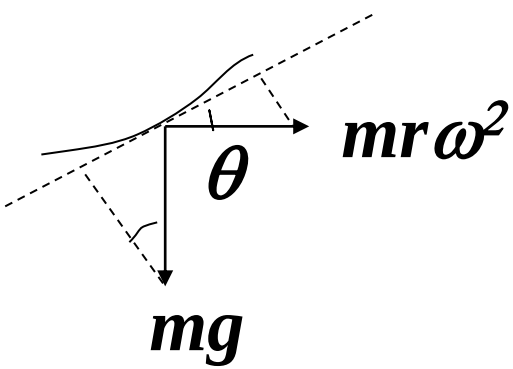
*在地表面用 ***g***，已考虑惯性离心力在内

这是基于流体的特性。流体不能承受切向力（剪切力），在静止或相对稳定的状态下，流体内部的压力作用方向总是垂直于作用面

例：水桶以ω旋转，求水面形状？



解：水面 z 轴对称，选柱坐标系。任选水面一小质元，在切线方向静止，在旋转参考系

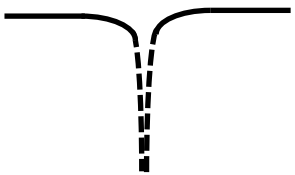


$$mg\sin\theta - mr\omega^2\cos\theta = 0$$

$$\tan\theta = \frac{r\omega^2}{g} \rightarrow \frac{dz}{dr} = \frac{r\omega^2}{g}$$

积分 $$\int_{z_0}^{z} dz = \int_0^r dr\frac{r\omega^2}{g} \quad -> \quad z = z_0 + \frac{r^2\omega^2}{2g}$$

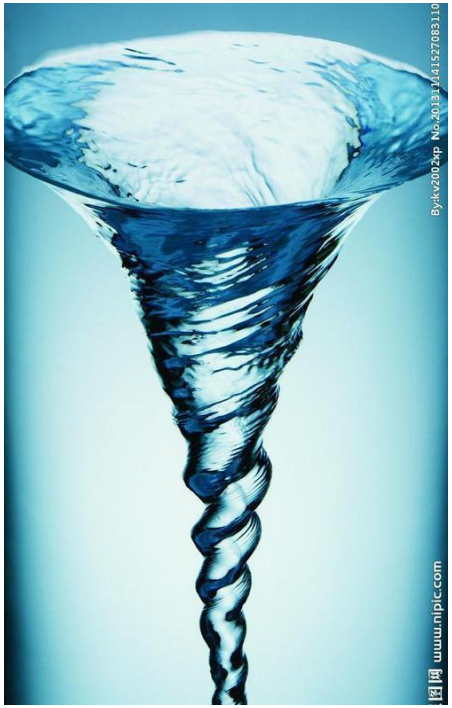
旋涡面



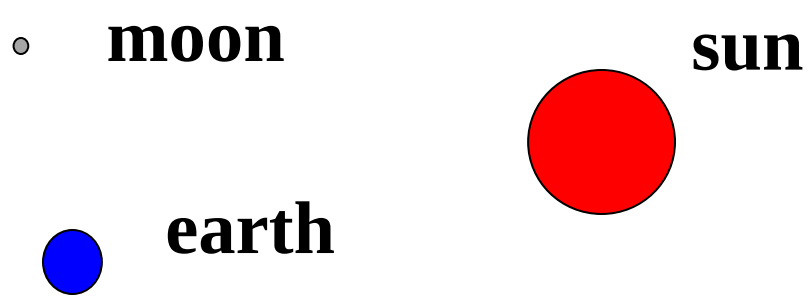
$$\oint \vec{\upsilon} \cdot d\vec{r} = \text{const.} \qquad 2\pi r \upsilon = const.$$

$$\frac{dz}{dr} = \frac{r\omega^2}{g} = \frac{c^2}{r^3 g}$$

$$z = z_0 - \frac{c^2}{2g}\frac{1}{r^2}$$



例：



$$\vec{F}_{E,M} \approx M_M \vec{a} \quad ?$$

求： 1. 月球所受地球和太阳的引力之比？

2. 月球在地心参考系的运动方程？

解：1.

$$\frac{F_{E,M}}{F_{S,M}} = \frac{\dfrac{M_E}{r_{E,M}^2}}{\dfrac{M_S}{r_{S,M}^2}} = \frac{r_{S,M}^2}{r_{E,M}^2}\frac{M_E}{M_S}$$

$$=\left(\frac{1.49\times 10^{11}\mathrm{m}}{3.84\times 10^{8}\mathrm{m}}\right)^2 \frac{5.975\times 10^{24}\mathrm{kg}}{1.99\times 10^{30}\mathrm{kg}} \approx 0.45$$

2. 在地心参考系, 月球 $\vec{F}_{E,M} + \vec{F}_{S,M} + \vec{F}_{inertia} = M_M \vec{a}$

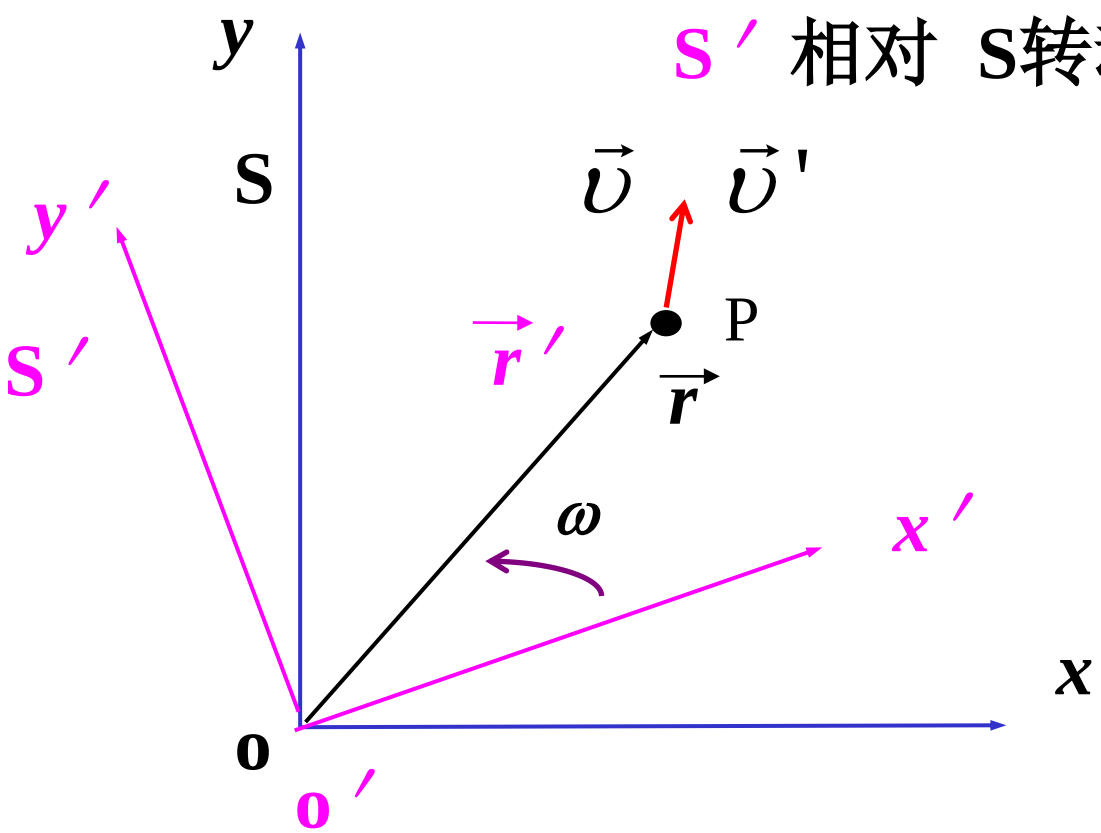
在太阳参考系地球受力 $\vec{F}_{M,E} + \vec{F}_{S,E} = M_E \vec{a}_0$

$$\frac{F_{M,E}}{F_{S,E}} \approx 0.0055 \qquad \vec{F}_{S,E} \approx M_E \vec{a}_0$$

$$\vec{F}_{S,M} + \vec{F}_{inertia} \approx \frac{M_M}{M_E} \vec{F}_{S,E} - M_M \vec{a}_0 \approx 0$$

$$\vec{F}_{E,M} \approx M_M \vec{a}$$

二、科氏加速度和科氏力*



S′ 相对 S转动，角速度为 $\vec{\omega}$ 常量

$$\vec{r} = \vec{r}\,'$$

$$\vec{\upsilon} = \vec{\upsilon}\,' + \vec{\omega} \times \vec{r}\,'$$

$$\vec{a} = \vec{a}\,' + \underbrace{2\vec{\omega} \times \vec{\upsilon}\,'}_{\text{科里奥利加速度}} + \underbrace{\vec{\omega} \times (\vec{\omega} \times \vec{r}\,')}_{\text{向心加速度}}$$

S′ 相对 S转动，角速度为 $\vec{\omega}$ 常量

$$\vec{a} = \vec{a}' + 2\vec{\omega}\times\vec{\upsilon}' + \vec{\omega}\times(\vec{\omega}\times\vec{r}')$$

结果对于三维也是正确的

S参考系

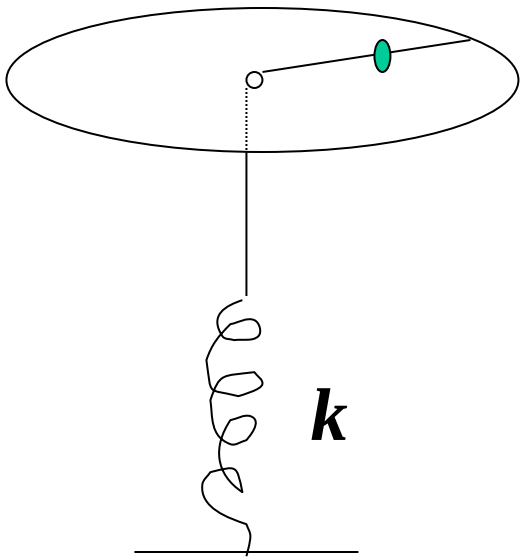
$$\vec{F} = m\vec{a} = m\vec{a}' + 2m\vec{\omega}\times\vec{\upsilon}' + m\vec{\omega}\times(\vec{\omega}\times\vec{r}')$$

S′ 系

$$\vec{F} + \underbrace{2m\vec{\upsilon}'\times\vec{\omega}}_{\text{科里奥利力}} \underbrace{- m\vec{\omega}\times(\vec{\omega}\times\vec{r}')}_{\text{离心力}} = m\vec{a}'$$

$$-m\vec{\omega}\times(\vec{\omega}\times\vec{r}') = m\omega^2\vec{r}$$

例. 桌面匀角速转动，一质点在桌面上的径向凹槽内, 作无摩擦匀速运动

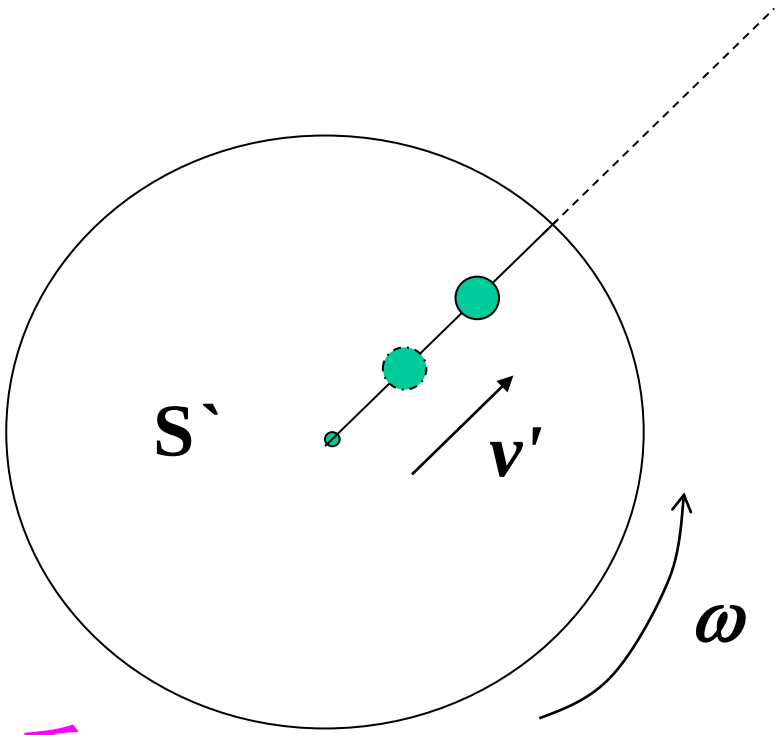


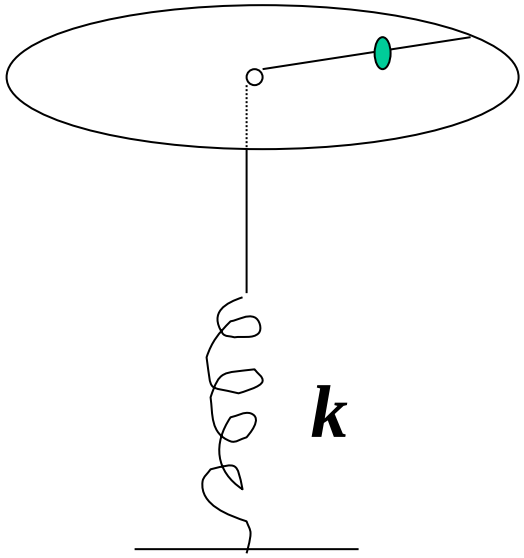
质点在圆心上 $r=0$，弹簧为自然长度，r 处

$$kr = mr\omega^2$$

$$k = m\omega^2$$

弹性力与离心力总处于平衡，在转动参考系一旦质点沿径向获速 v'，将保持匀速直线运动。





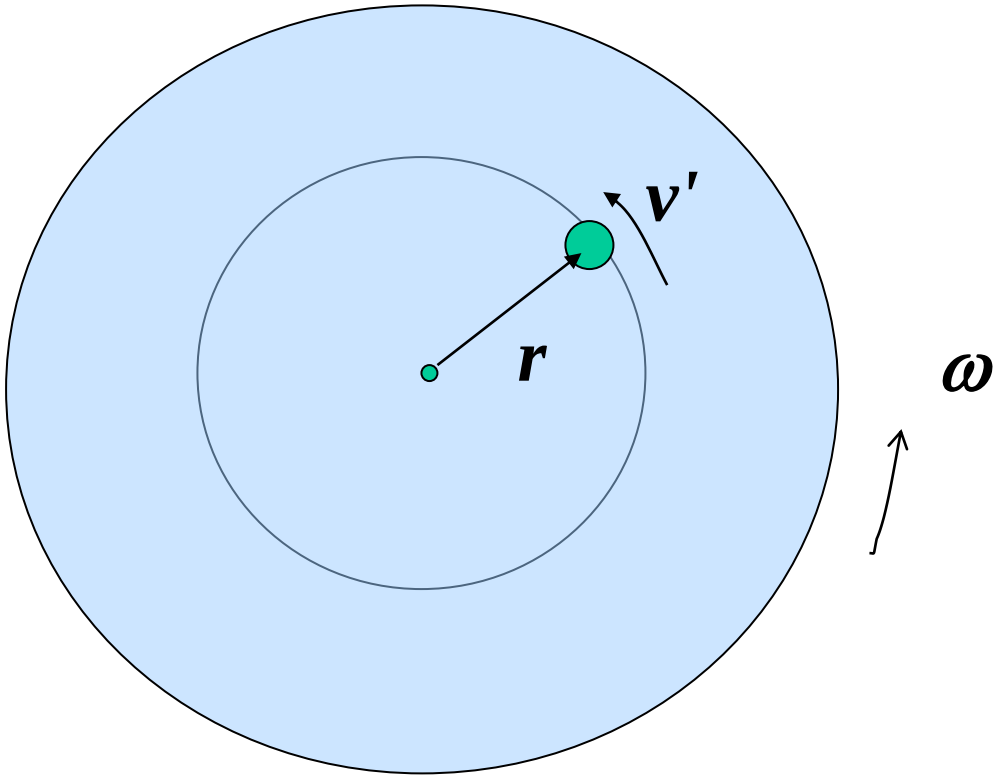
S′系

$$\vec{F}+2m\vec{\upsilon}'\times\vec{\omega}-m\vec{\omega}\times(\vec{\omega}\times\vec{r}')=m\vec{a}'=0$$

科氏力与槽壁支持力平衡

离心力与弹簧拉力平衡

例. 桌面匀角速转动，一质点在桌面上的同心圆环凹槽内，作无摩擦匀速运动



槽壁支持力 $f\hat{n}$

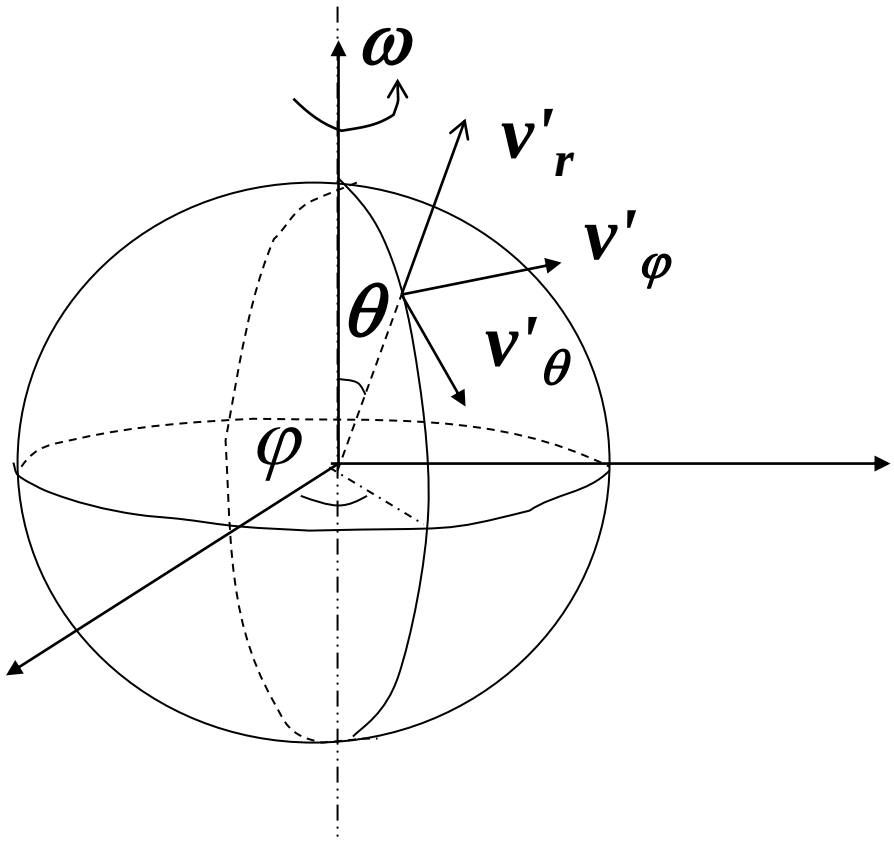
S′系

$$\vec{F}+2m\vec{\upsilon}'\times\vec{\omega}-m\vec{\omega}\times(\vec{\omega}\times\vec{r}\,')=m\vec{a}\,'$$

$$f\hat{n}+2m\vec{\upsilon}'\times\vec{\omega}-m\vec{\omega}\times(\vec{\omega}\times\vec{r}\,')=m\frac{\upsilon'^2}{r'}\hat{n}$$

槽壁支持力+科氏力+离心力 共同提供向心加速度

3. 推广到三维球面

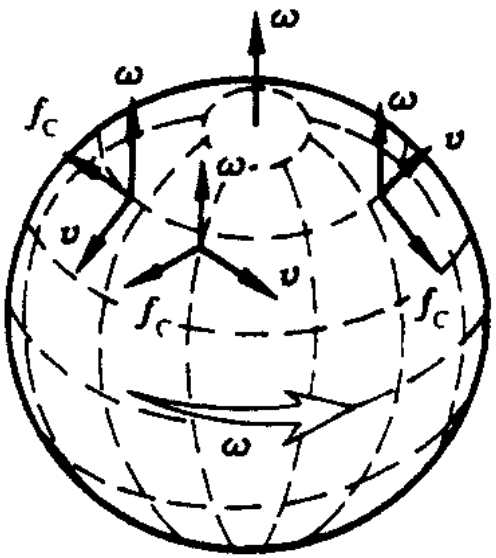


$$(\hat{r}, \hat{\theta}, \hat{\varphi})$$

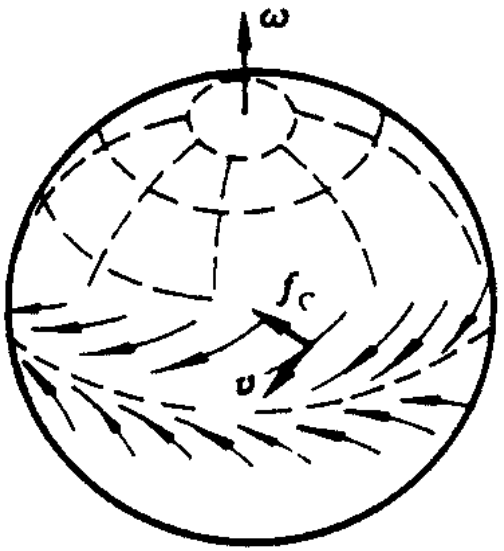
$$\vec{f}_c = 2m\vec{v}\,' \times \vec{\omega}$$

落体偏东

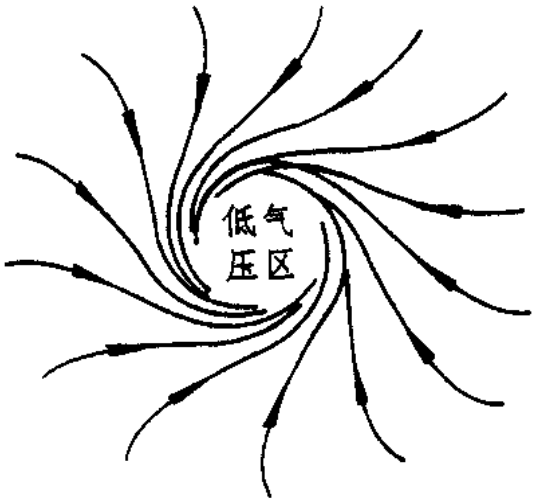
江水冲刷右岸



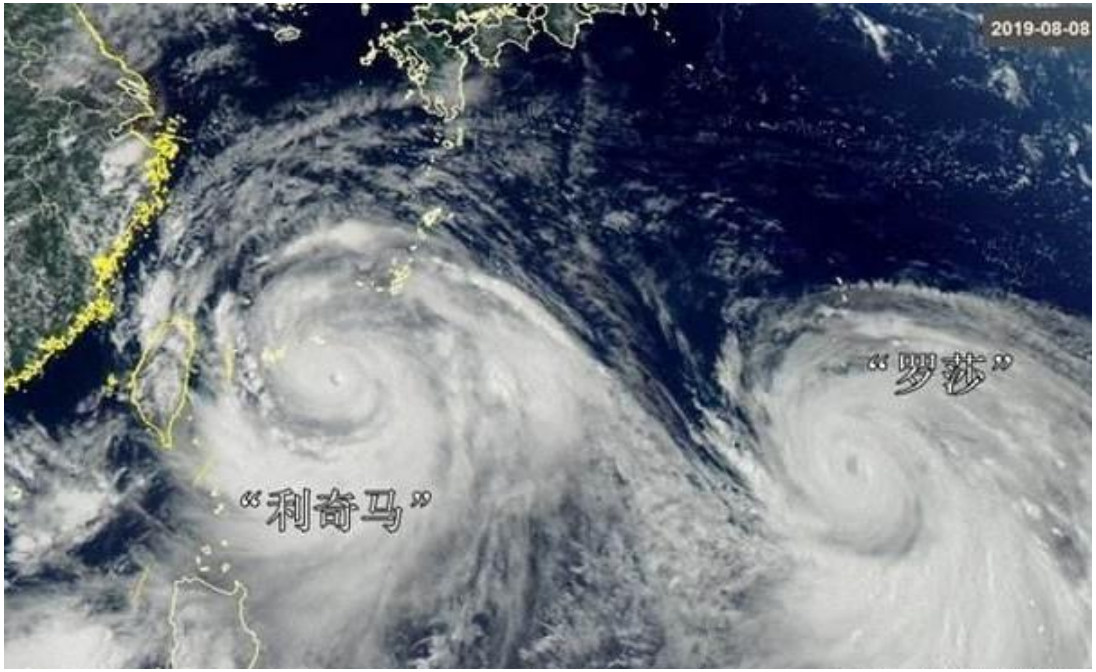
北半球上的科里奥利力

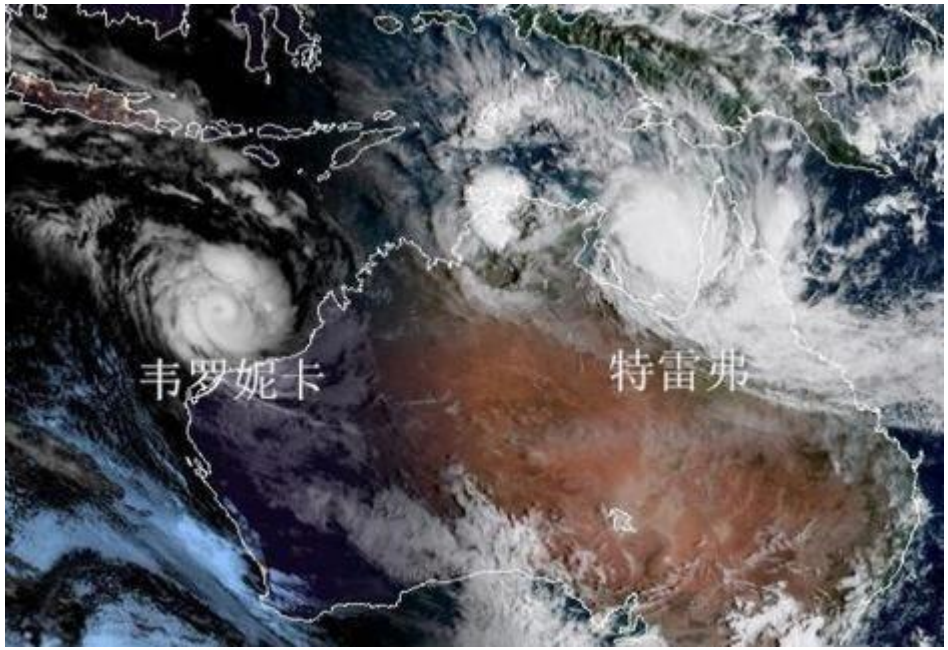


信风的形成

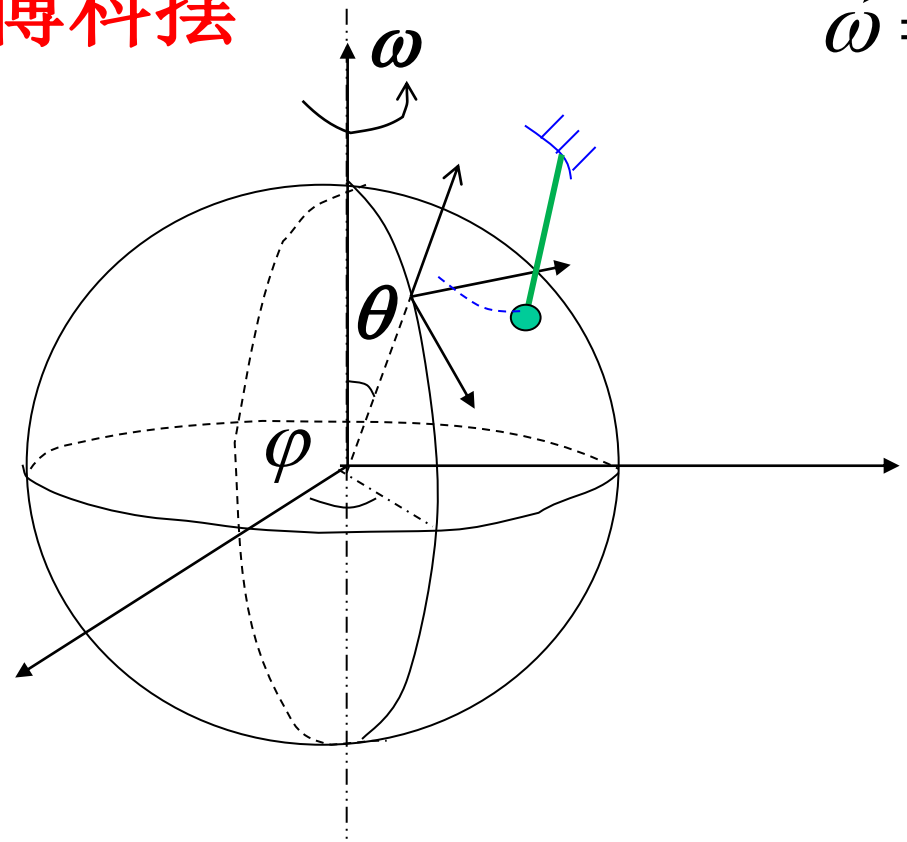


旋风的形成


2019-08-08
“罗莎”
“利奇马”


韦罗妮卡
特雷弗

傅科摆



$$\vec{\omega} = \omega_{\perp}\hat{r} + \omega_{//}\hat{\theta}$$

$$= \omega\cos\theta\hat{r} - \omega\sin\theta\hat{\theta}$$

地面上科氏力

$$\vec{f}_c = 2m\vec{\upsilon}' \times \vec{\omega}$$

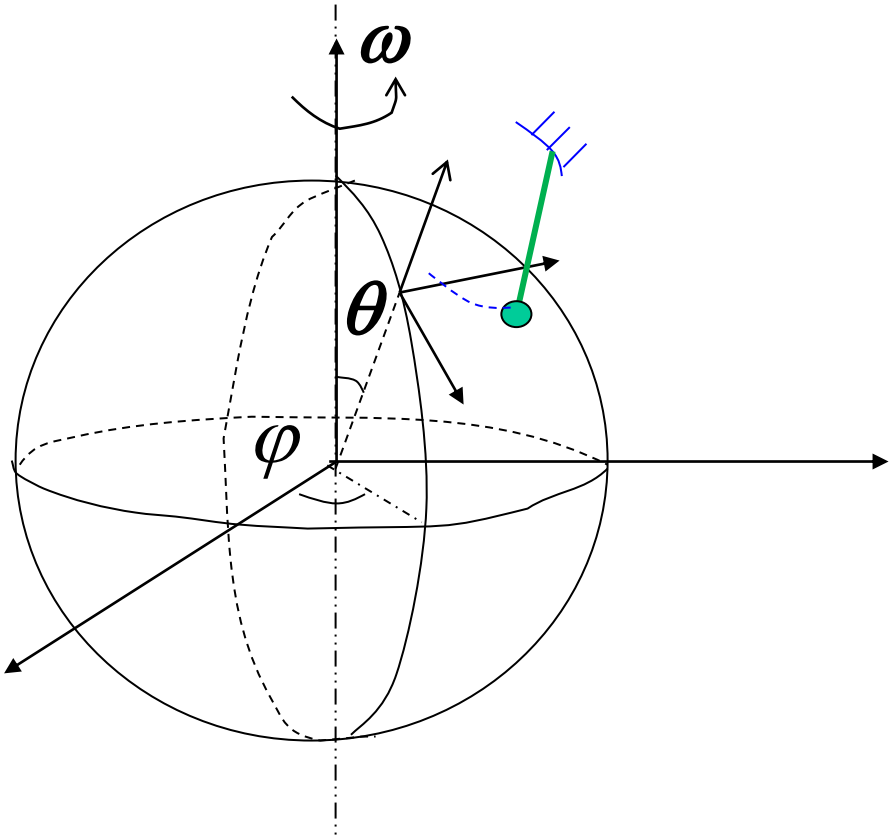
$$\vec{\upsilon}' = \upsilon'_{\perp}\hat{r} + \upsilon'_{//}\hat{t}$$

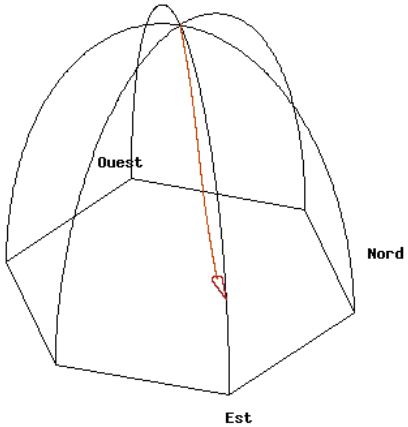
$$\approx \upsilon'\hat{t}$$

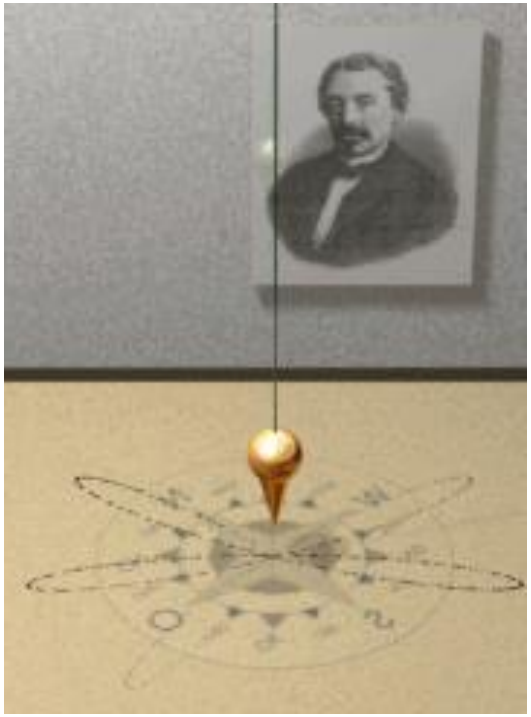
科氏力 $\underline{2m\upsilon'\omega_{\perp}\hat{t}\times\hat{r}} + 2m\upsilon'\omega_{//}\hat{t}\times\hat{\theta}$

横向加速度 横向 径向

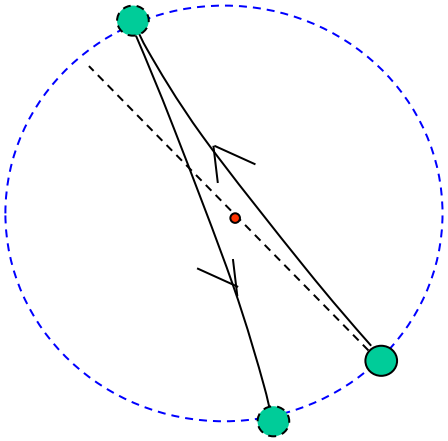
$a_c = 2\upsilon'\omega_{\perp}$ 使摆平面旋转 不影响摆平面





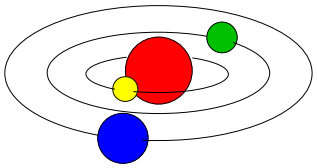


$\omega_{\perp}$



科氏力

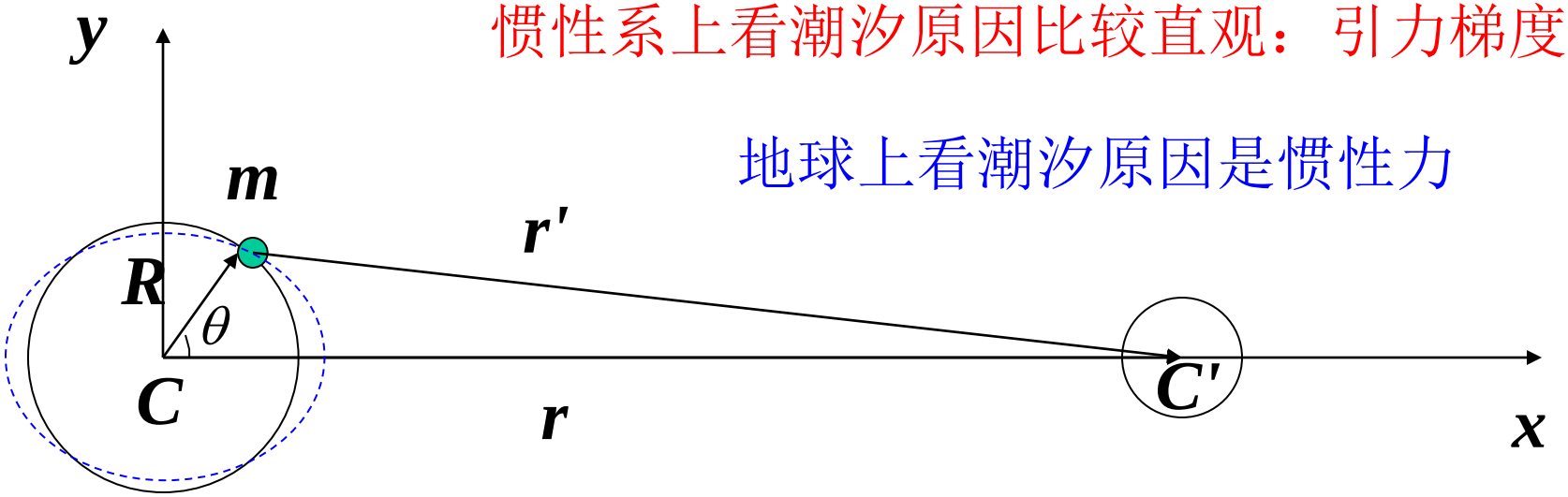
$$2m\upsilon'\omega_{\perp}\hat{\upsilon}'\times\hat{r}$$



§ 2.9 潮汐力和潮汐(tide)*

惯性系上看潮汐原因比较直观：引力梯度

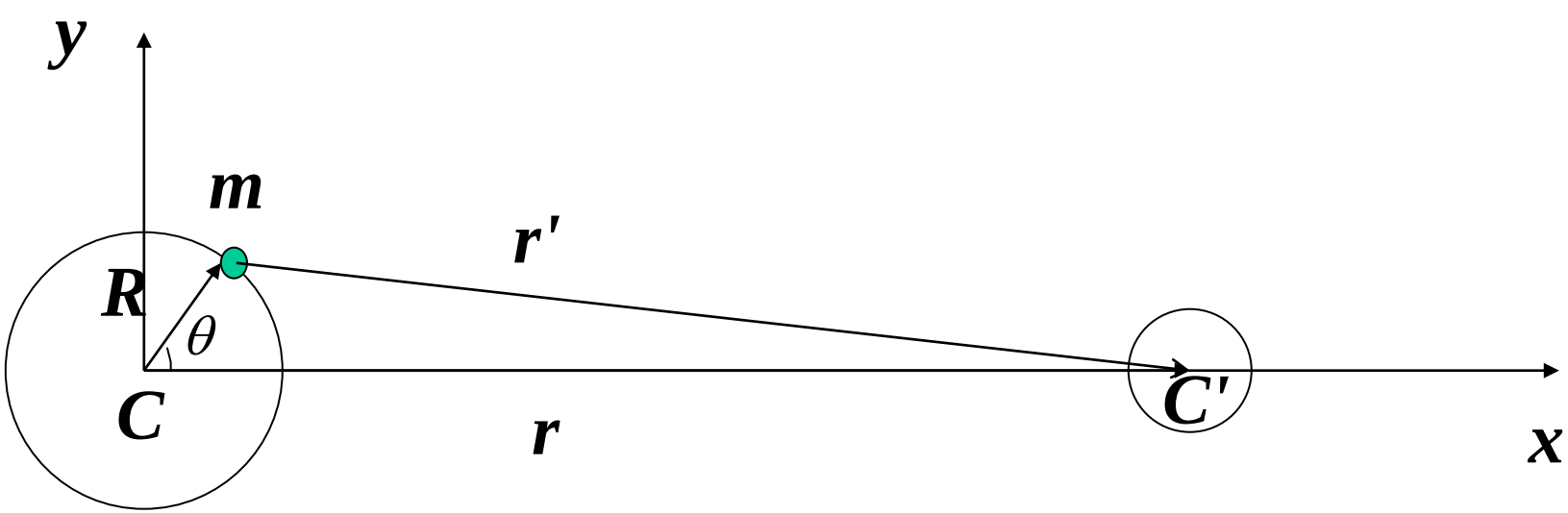
地球上看潮汐原因是惯性力



一小块海水受月球引力 $\vec{f}_M = G\dfrac{mM_M}{r'^2}\hat{r}'$

地心运动

$$\vec{f}_{E,M} = G\frac{M_E M_M}{r^2}\hat{r} = M_E\vec{a}_0$$



在地心参考系

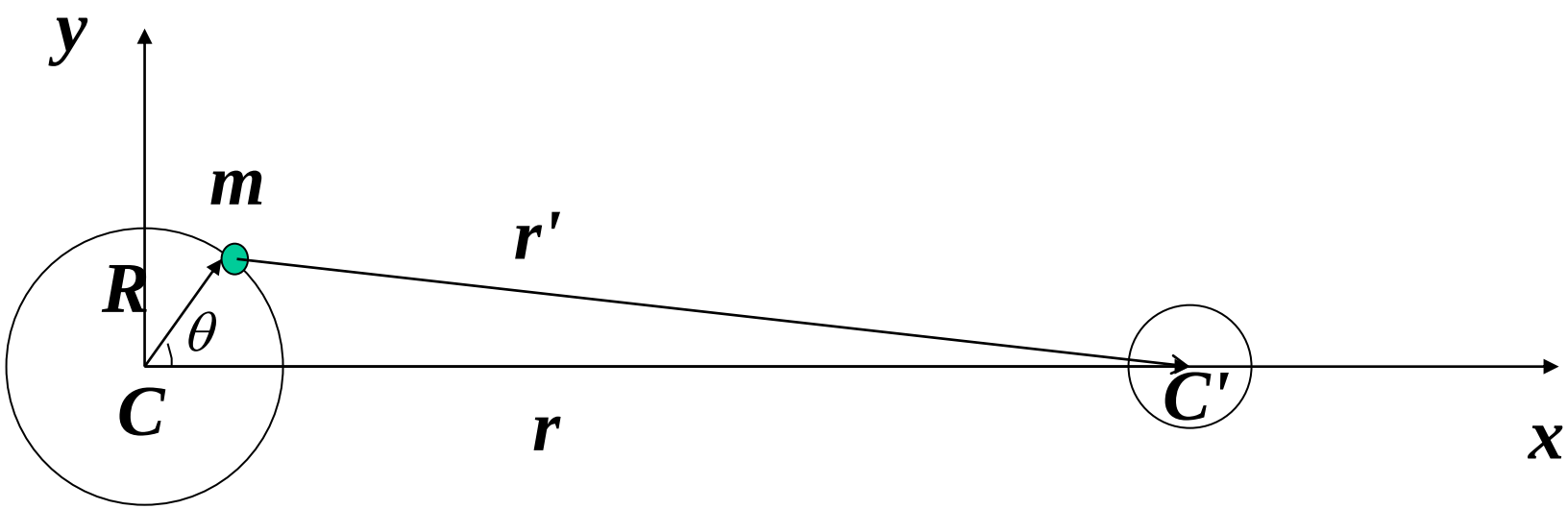
$$\vec{f}_E + \vec{f}_{物质间作用} + \vec{f}_M + \vec{f}_{惯} = m\vec{a}$$

地心参考系下
剩余力

$$\vec{f}_M + \vec{f}_{惯} = \vec{f}_{潮汐}$$

$$\vec{f}_{惯} = -m\vec{a}_0 = -G\frac{mM_M}{r^2}\hat{r} \qquad \vec{f}_M = G\frac{mM_M}{r'^2}\hat{r}'$$

$$\vec{f}_{潮汐} = GmM_M\left(\frac{\hat{r}'}{r'^2} - \frac{\hat{r}}{r^2}\right) = GmM_M\left(\frac{\vec{r}'}{r'^3} - \frac{\vec{r}}{r^3}\right)$$

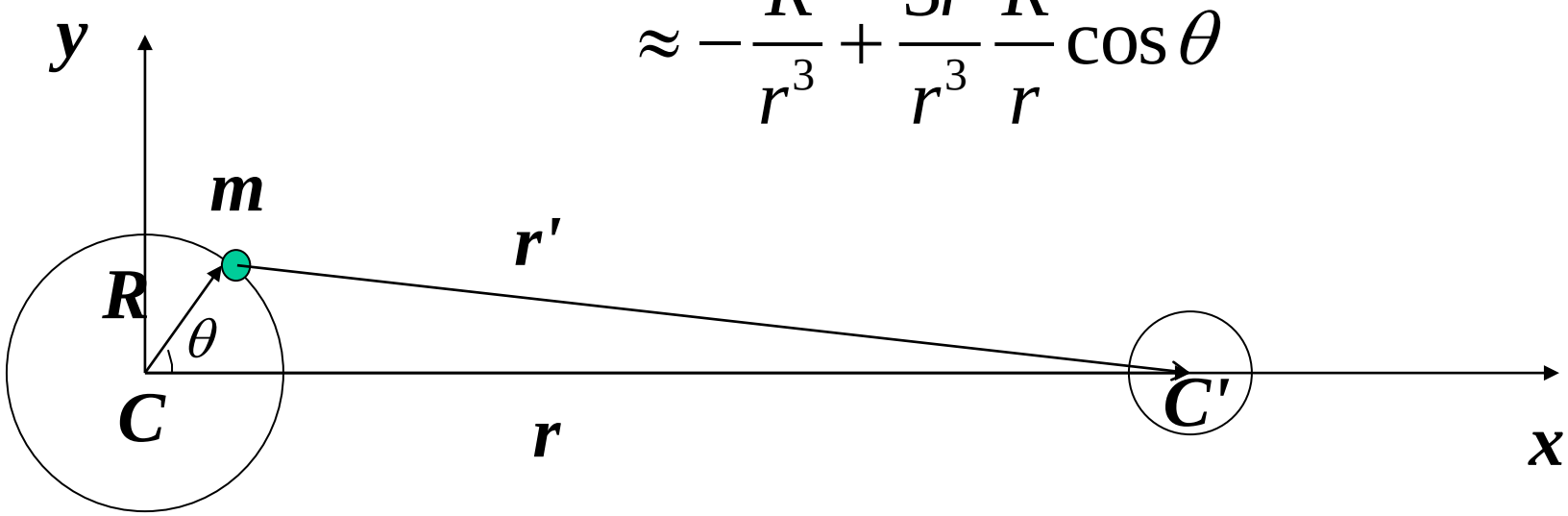


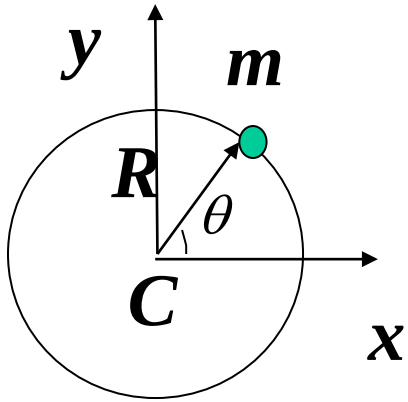
$$r'^{2} = r^{2} + R^{2} - 2Rr\cos\theta$$

$$r >> R \qquad \frac{1}{r'^{3}} = \frac{1}{r^{3}}\left(1 + 3\frac{R}{r}\cos\theta\right)$$

$$\frac{\vec{r}\,'}{r'^{3}} - \frac{\vec{r}}{r^{3}} \approx \frac{\vec{r} - \vec{R}}{r^{3}}\left(1 + 3\frac{R}{r}\cos\theta\right) - \frac{\vec{r}}{r^{3}}$$

$$\approx -\frac{\vec{R}}{r^{3}} + \frac{3\vec{r}}{r^{3}}\frac{R}{r}\cos\theta$$

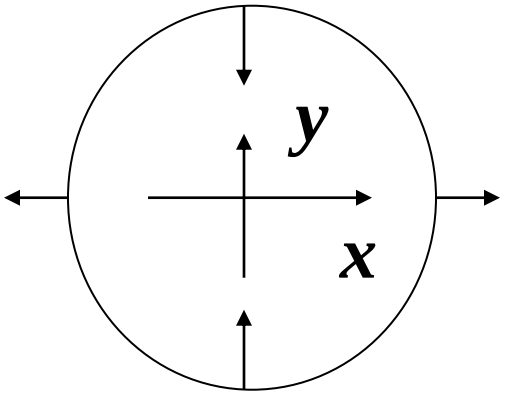




$$\vec{f}_{tide} = \frac{GmM_M}{r^3}\left(-\vec{R} + \frac{3\vec{r}}{r}R\cos\theta\right)$$

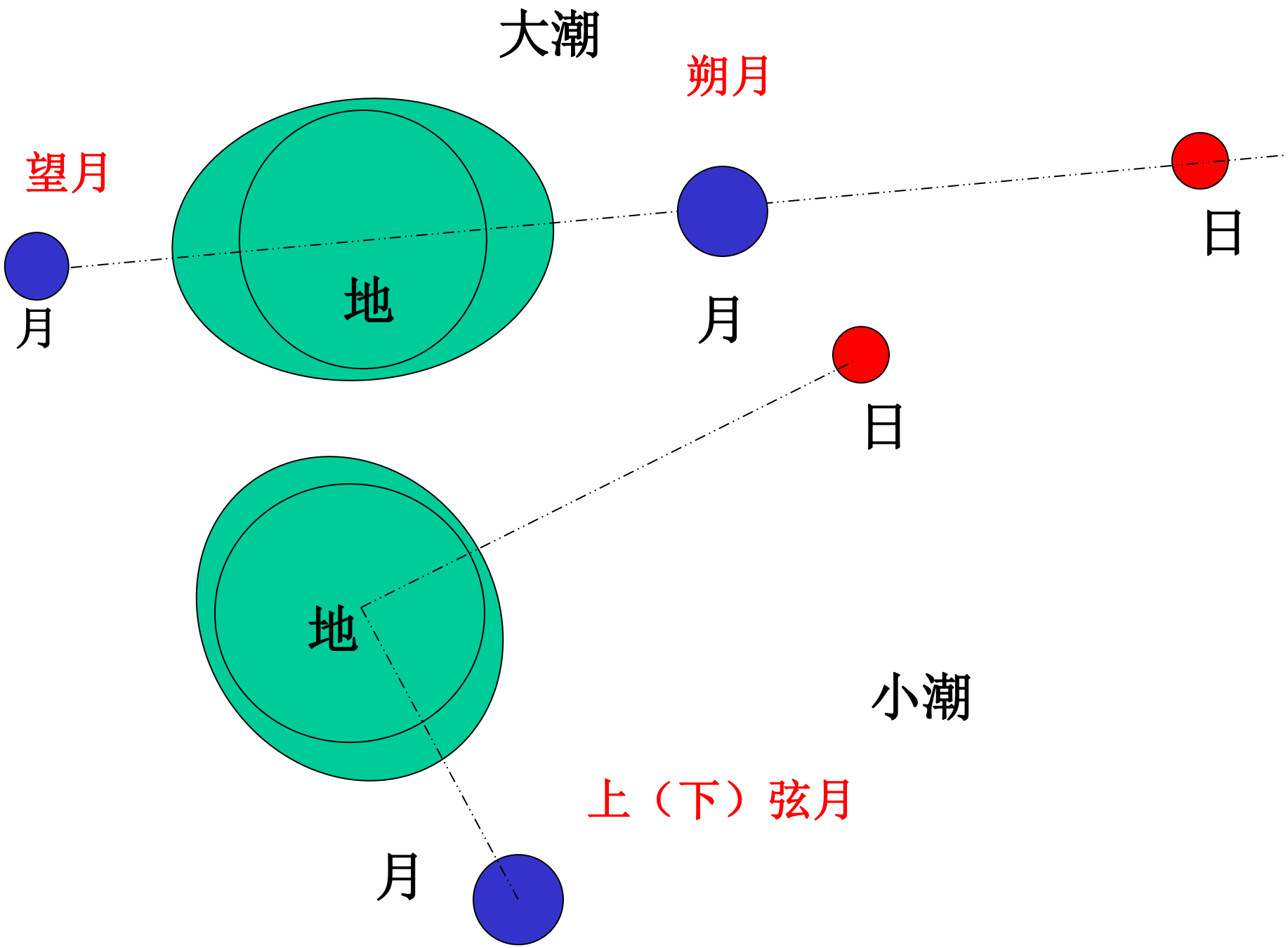
***r* 地月间距**

$$(f_{潮汐})_x \approx 2GmM_M \frac{R\cos\theta}{r^3}$$

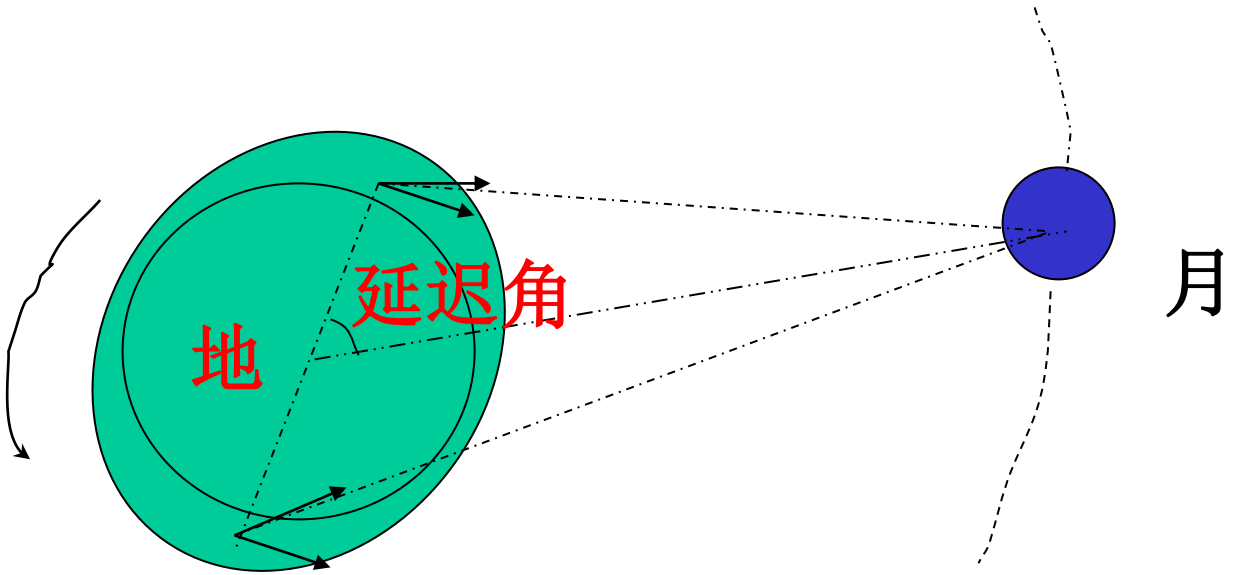


$$(f_{潮汐})_y \approx -GmM_M \frac{R\sin\theta}{r^3}$$

$$\frac{(f_{潮汐})_{月}}{(f_{潮汐})_{日}} = \frac{M_M}{M_S}\left(\frac{r_{E,S}}{r_{E,M}}\right)^3 \approx 2.18$$


大潮
朔月
望月
日
月
地
月
日
地
小潮
上（下）弦月
月

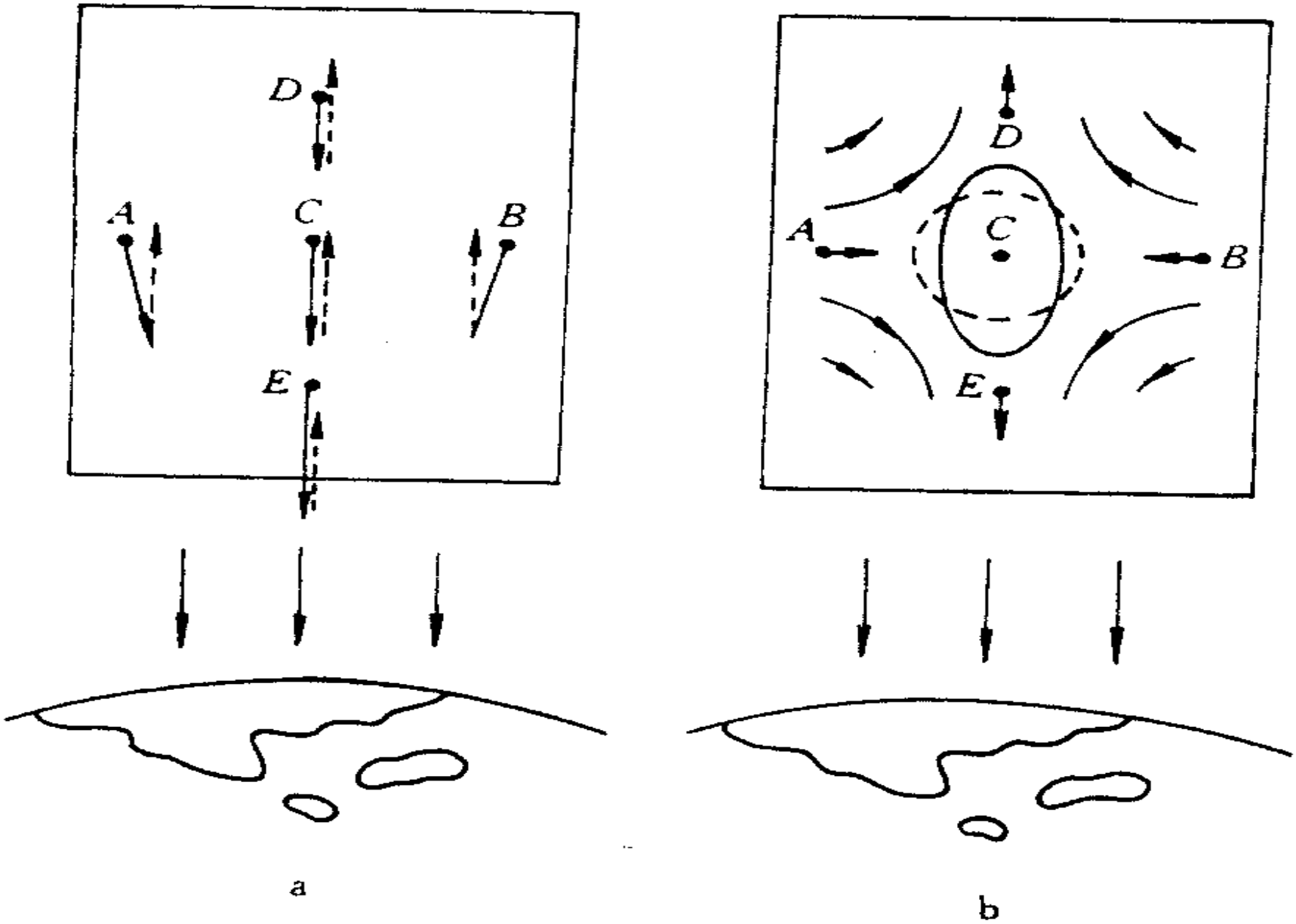
固体潮：引潮力对固体作用使之形变，但应变稍有延迟



英国的开尔文（Kelvin）(1876)，达尔文（G.H.Darwin）(1883)

月球在地球上引起的固体潮形成阻止地球自转的反力矩，减慢地球自转速度，3亿年前地球400天/每年，现只有365.25天/每年.

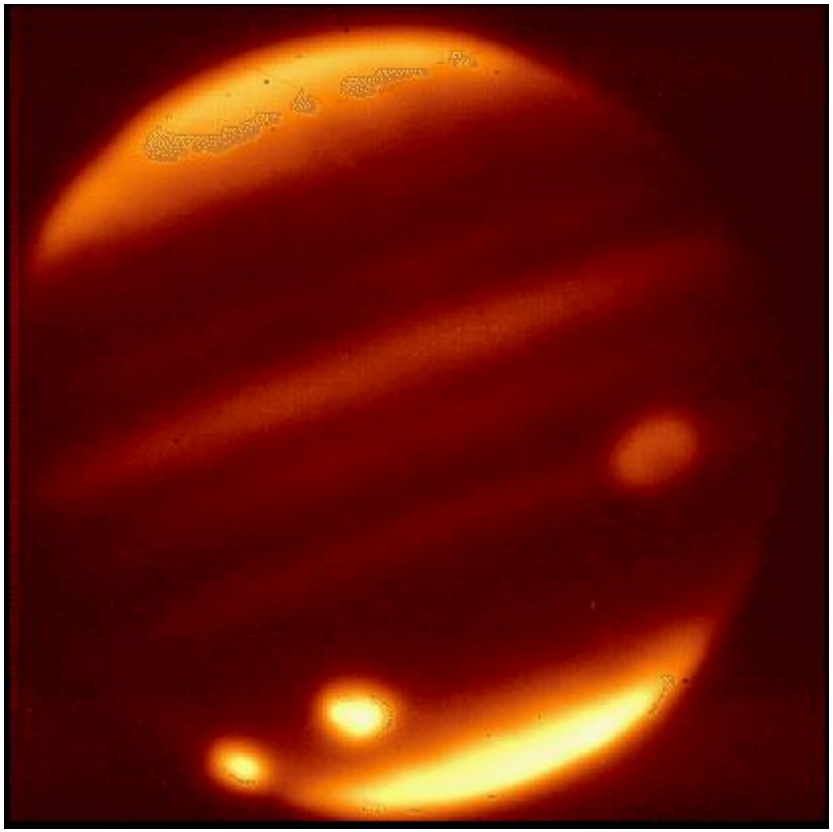
地对月：月球自转和公转周期相同



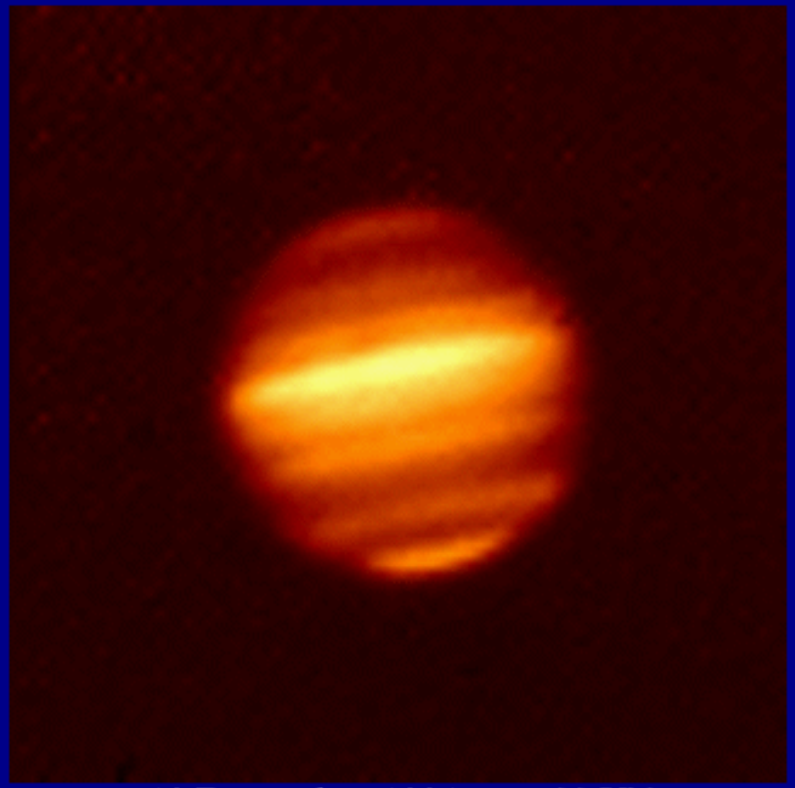
自由降落升降机中的引潮力

月球撞击地球前，引潮力将撕碎它

彗星撞击木星

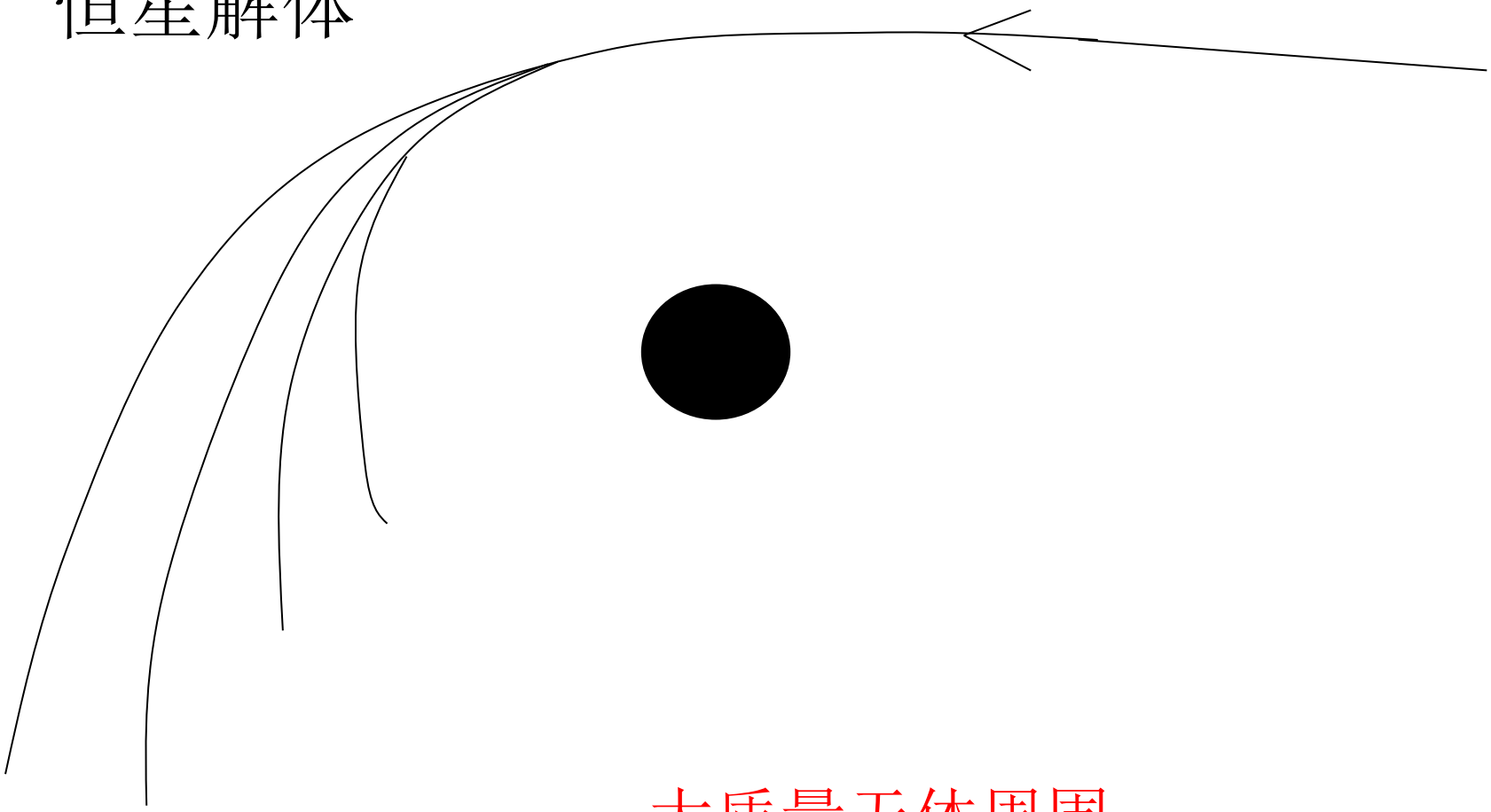


Jupiter Impact Effects Still Visible



19 December 1994 • 7:00 UT
Calar Alto 3.5 m Telescope + MAGIC Infrared Camera
1.7 μm Methane Filter, showing continuous band
in south mid–latitudes due to SL-9 impacts

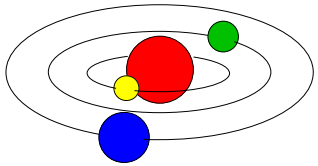
恒星解体



大质量天体周围
恒星受引潮力

地震与潮汐的关系：引潮力触发地震

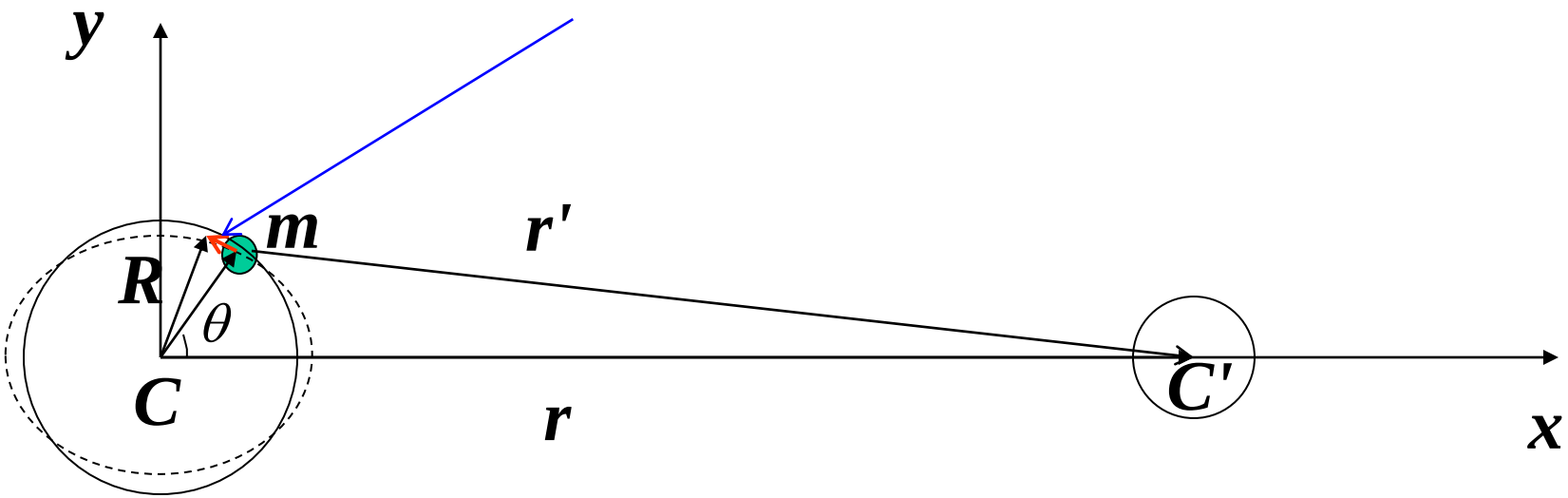
①常发生于夜间
②常发生于阴历初一，十五（大潮期） 左右。
唐山地震：76农七月初二
神户地震：95农十二月十七
印度地震：93农八月十五



潮汐高度*

$$\vec{f}_{潮汐} = \frac{GmM_M}{r^3}\left(-\vec{R} + \frac{3\vec{r}}{r}R\cos\theta\right)$$

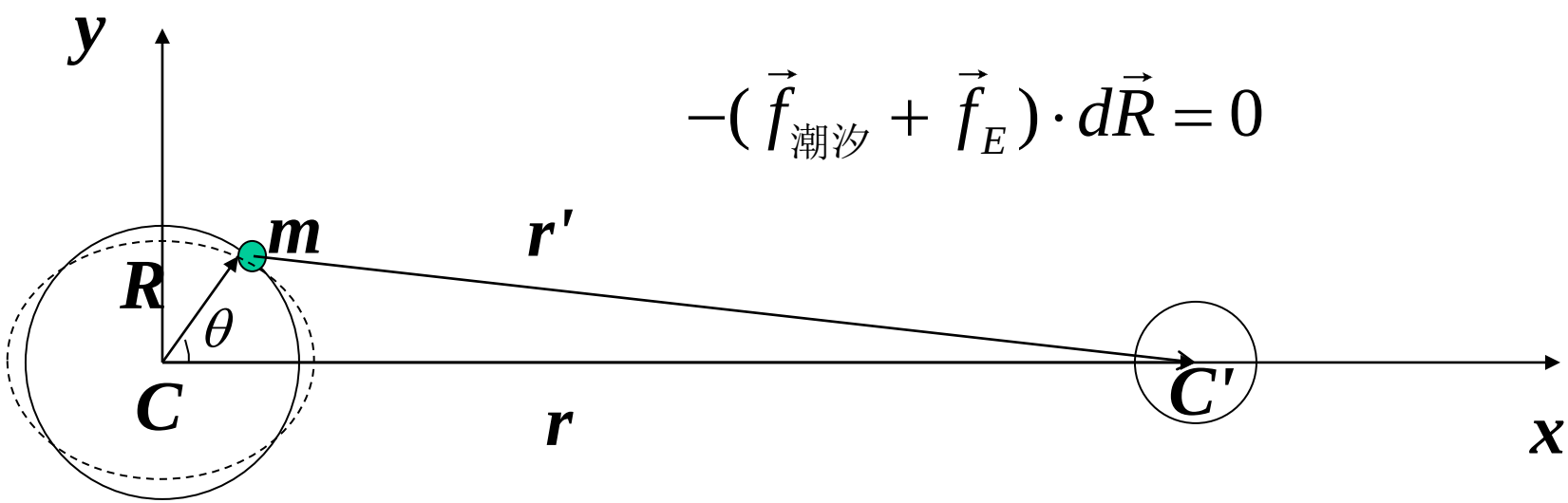
$$d\vec{R} = \hat{R}dR + Rd\theta\hat{\theta}$$



沿等势面(虚线) 移动不做功

$$dU = 0 = -(\vec{f}_{潮汐} + \vec{f}_E)\cdot d\vec{R}$$

$$\vec{f}_E = G\frac{mM_E}{R^3}\vec{R}$$



$$-(\vec{f}_{潮汐}+\vec{f}_E)\cdot d\vec{R}=0$$

$$\frac{M_M}{r^3}(-RdR+3RdR\cos^2\theta-3R^2\cos\theta\sin\theta d\theta)+\frac{M_E}{R^2}dR$$

$$\frac{M_E}{R^2}>>\frac{M_M R}{r^3}$$

$$-3R^2\frac{M_M}{r^3}\cos\theta\sin\theta d\theta+\frac{M_E}{R^2}dR=0$$

$$\Delta U = 0 = -\int_{\theta=0}^{\theta} (\vec{f}_{潮汐} + \vec{f}_E) \cdot d\vec{R}$$

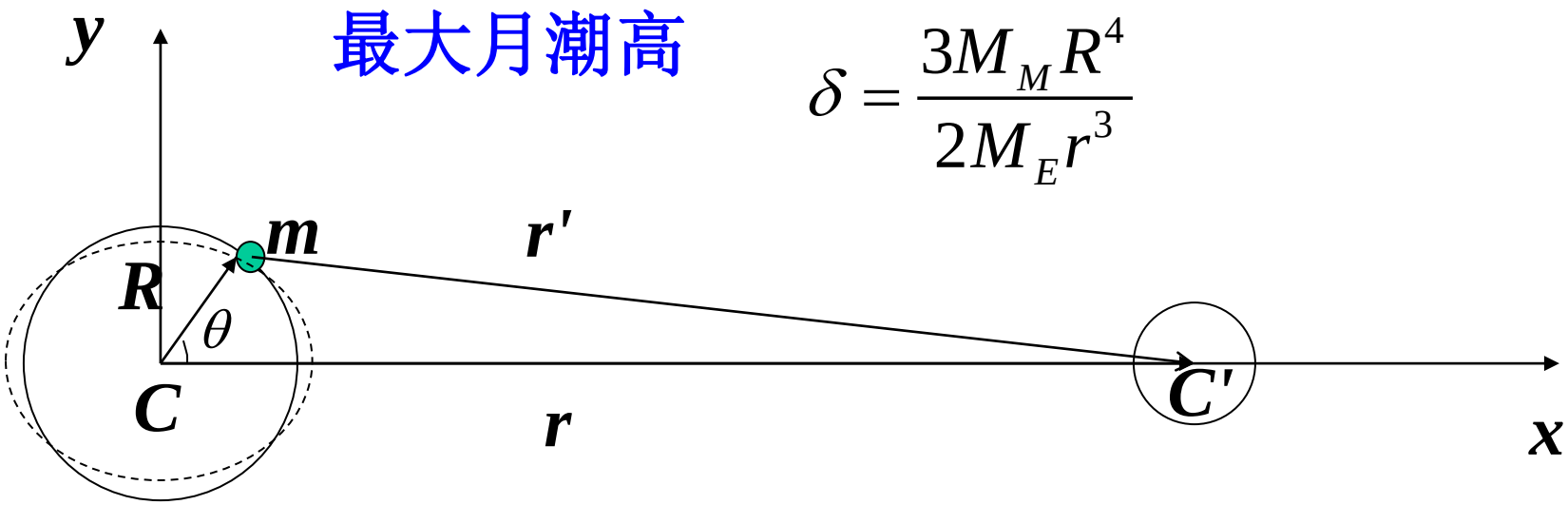
$$\int_0^{\theta} \frac{dR}{R^2} \approx -\frac{\delta}{R^2}$$

设潮高 $\delta = R_{\theta=0} - R_\theta$

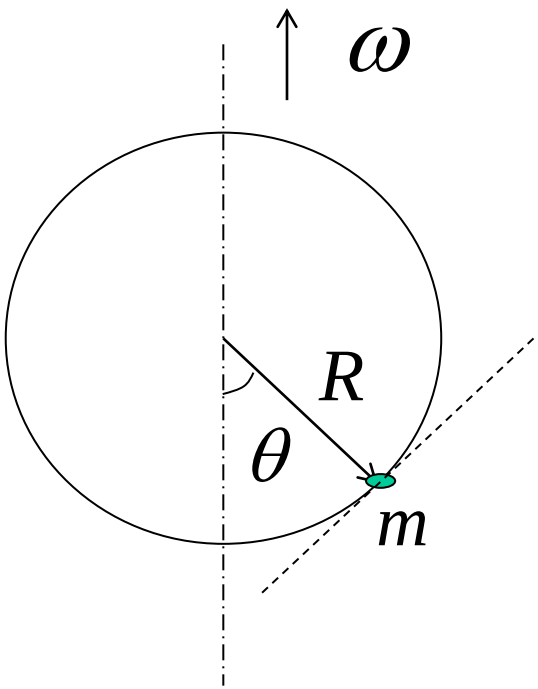
$$\frac{3M_M}{2r^3}\sin^2\theta = \delta\frac{M_E}{R^4}$$

最大月潮高

$$\delta = \frac{3M_M R^4}{2M_E r^3}$$



稳定平衡



圆环绕对称轴（直径）匀速转动

圆环上套一质点，可自由滑动

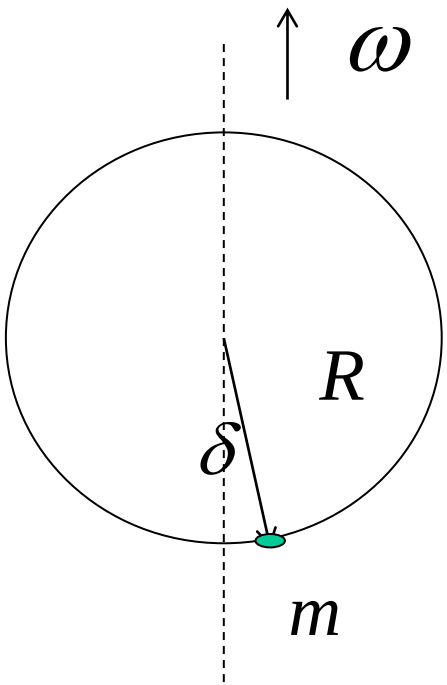
找到平衡点，判断是否为稳定

$$F_\theta = m\omega^2 R\sin\theta\cos\theta - mg\sin\theta = 0$$

平衡点 $\sin\theta_0 = 0$

$$\theta_0 = 0, \quad \pi$$

$$\omega^2 > \frac{g}{R} \qquad \cos\theta_0 = \frac{g}{\omega^2 R}$$



$$F_\theta = m\omega^2 R\sin\theta\cos\theta - mg\sin\theta = 0$$

$\theta_0 = 0$ 扰动 $\theta = \theta_0 + \delta = \delta$

$$F_\theta = m(\omega^2 R\cos\delta - g)\delta$$

$$\omega^2 \le \frac{g}{R}$$

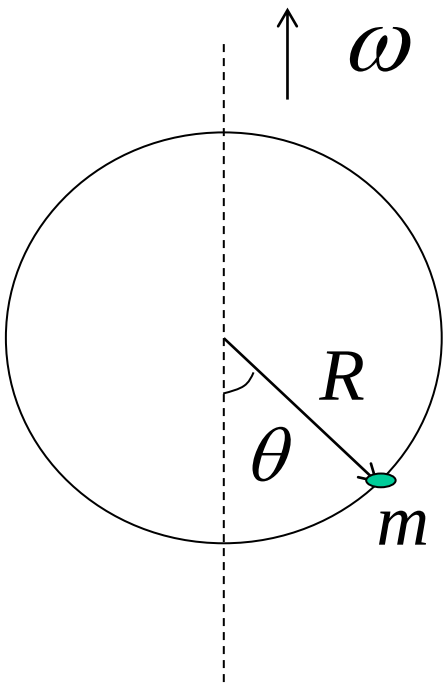
$$F_\theta \propto -\delta$$

稳定

$$\omega^2 > \frac{g}{R}$$

$$F_\theta \propto \delta$$

不稳定



$$\theta_0 = \pi \qquad \theta = \theta_0 + \delta = \pi + \delta$$

$$F_\theta = m(\omega^2 R\cos\delta + g)\delta \propto \delta$$

不稳定

$$\omega^2 > \frac{g}{R} \qquad \cos\theta_0 = \frac{g}{\omega^2 R}$$

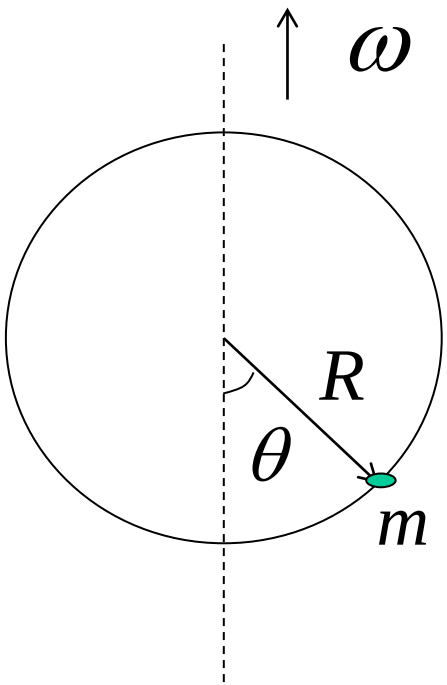
$$\theta = \theta_0 + \delta \qquad F_\theta = m\omega^2 R\sin\theta\cos\theta - mg\sin\theta = 0$$

$$\cos(\theta_0 + \delta) = \cos\theta_0\cos\delta - \sin\theta_0\sin\delta = \cos\theta_0 - \delta\sin\theta_0$$

$$F_\theta = m[\omega^2 R\cos(\theta_0 + \delta) - g]\sin(\theta_0 + \delta) = -m\omega^2 R\delta\sin^2\theta_0$$

$$F_\theta \propto -\delta$$

稳定



$$F_\theta = m\omega^2 R\sin\theta\cos\theta - mg\sin\theta$$

稳定性

$$F'|_{\theta=\theta_0} \quad \begin{matrix} >0 & \text{不稳定} \\ <0 & \text{稳定} \end{matrix}$$

$$F'|_{\theta=\theta_0} = m\omega^2 R\cos 2\theta_0 - mg\cos\theta_0$$

$$\theta_0 = 0, \quad \pi$$

若等于0，还要考察更高一级导数

$$\omega^2 > \frac{g}{R} \qquad \cos\theta_0 = \frac{g}{\omega^2 R}$$

第二章完

本章编者

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